

ISTQB Foundation Sample Exam Questions

***Certified Tester Foundation Level
(CTFL) 2018 Syllabus***

2nd Edition

(Updated for CTFL Syllabus 2018 V3.1)

Chhavi Raj Dosaj

ISTQB certified registered trainer for ISTQB Foundation Level
(CTFL), ISTQB Foundation Level- Agile Tester Extension
(CTFL-AT) & ISTQB Advanced Level- Test Manager
(CTFL-ATM)

ISTQB Foundation Sample Exam Questions - Certified Tester Foundation Level (CTFL) 2018 Syllabus

By Chhavi Raj Dosaj

Copyright © Chhavi Raj Dosaj

No part of this publication may be reproduced, stored in a retrieval system or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, scanning, or otherwise without either the prior written permission of the Author or authorization through payment of the appropriate per-copy fee to the Author. For permission please contact the author.

This document also contains registered trademarks, trademarks and service marks that are owned by their respective companies or organizations. The Publisher and the author disclaim any responsibility for specifying which marks are owned by which companies or organizations.

LIMIT OF LIABILITY/DISCLAIMER OF WARRANTY: THE PUBLISHER AND THE AUTHOR MAKE NO REPRESENTATIONS OR WARRANTIES WITH RESPECT TO THE ACCURACY OR COMPLETENESS OF THE CONTENTS OF THIS WORK AND SPECIFICALLY DISCLAIM ALL WARRANTIES, INCLUDING WITHOUT LIMITATION, WARRANTIES OF FITNESS FOR A PARTICULAR PURPOSE. NO WARRANTY MAY BE CREATED OR EXTENDED BY SALES OR PROMOTIONAL MATERIALS. THE ADVICE AND STRATEGIES CONTAINED HEREIN MAY NOT BE SUITABLE FOR EVERY SITUATION. THIS WORK IS SOLD WITH THE UNDERSTANDING THAT THE PUBLISHER IS NOT ENGAGED IN RENDERING LEGAL, ACCOUNTING OR OTHER PROFESSIONAL SERVICES. IF PROFESSIONAL ASSISTANCE IS REQUIRED, THE SERVICES OF A COMPETENT PROFESSIONAL PERSON SHOULD BE SOUGHT. NEITHER THE PUBLISHER NOR THE AUTHOR SHALL BE LIABLE FOR DAMAGES ARISING HEREFROM. THE FACT THAT AN ORGANIZATION OR WEBSITE IS REFERRED TO IN THIS WORK AS A CITATION AND/OR A POTENTIAL SOURCE OF FURTHER INFORMATION DOES NOT MEAN THAT THE AUTHOR OR THE PUBLISHER ENDORSES THE INFORMATION. THE ORGANIZATION OR WEBSITE MAY PROVIDE OR MAKE THEIR OWN RECOMMENDATIONS.

Dedicated to,

My parents Dr. Pradeep Dosaj & Mrs. Kaushal Dosaj

Contents

PREFACE.....	vii
1 OVERVIEW OF EXAM.....	1
2 PRACTICE EXAM 1.....	3
3 ANSWERS- PRACTICE EXAM 1.....	27
4 PRACTICE EXAM 2.....	65
5 ANSWERS - PRACTICE EXAM 2.....	91
6 PRACTICE EXAM 3.....	133
7 ANSWERS- PRACTICE EXAM 3.....	154

Preface

I wrote this book for people who are planning to take the ISTQB foundation certification exam based on the new 2018 syllabus. The readers should first go through the ISTQB Certified Tester- Foundation Level syllabus 2018 version and then try the practice exams to check their understanding about the subject. These practice exams can be used to find areas in need of further study.

These practice exams will help you to check your progress throughout your exam preparation and will provide confidence to face the real exam.

This book can also be used by people who want to test their knowledge of software testing based on the ISTQB 2018 syllabus.

Disclaimer

Although all efforts have been made to ensure the accuracy of the contents of this book, we cannot guarantee 100% correctness of the information contained herein.

If you find any factual anomalies, grammar or spelling errors please send it along with your comments and suggestions to the author. Good luck for the exam.

Acknowledgments

I would like to acknowledge that this book relies heavily on the new ISTQB Certified Tester- Foundation Level syllabus 2018 version and Standard Glossary of Terms used in Software Testing. In some cases, certain phrases were used verbatim to ensure the content adheres to the syllabus and glossary.

I would like to thank my wife Shikha for her support when I was busy writing this book.

Special thanks to Arun Kumar for helping me in the preparation of diagrams and Sona Rao for helping me with the technical review of the book.

I would also like to thank Sapna Bhatia and Navneesh Garg for their inspiration and support.

1

Overview of Exam

Introduction to Certified Tester Foundation Level

The Foundation Level qualification is aimed at anyone involved in software testing. This includes people in roles such as testers, test analysts, test engineers, test consultants, test managers, user acceptance testers, and software developers. This Foundation Level qualification is also appropriate for anyone who wants a basic understanding of software testing, such as product owners, project managers, quality managers, software development managers, business analysts, IT directors, and management consultants. Holders of the Foundation Certificate will be able to go on to higher-level software testing qualifications.

Learning Objectives

The syllabus categorizes learning objectives into three cognitive levels:

- **K1:** remember, recognize, and recall
- **K2:** understand, explain, give reasons, compare, classify and summarize
- **K3:** apply

The relevant learning objectives at K1, K2, and K3 levels are provided at the beginning of each chapter within each particular extension syllabus.

The distribution of the question based on the chapters and cognitive levels is as below:

	K1	K2	K3	Total
Chapter 1	2	6	0	8
Chapter 2	1	4	0	5
Chapter 3	1	3	1	5
Chapter 4	1	5	5	11
Chapter 5	2	5	2	9
Chapter 6	1	1	0	2
Total	8	24	8	40

Exam Structure

This exam is comprised of 40 multiple choice questions. To pass the exam, at least 65% of the questions (26 questions) must be answered correctly.

How to use this book for exam preparation

The readers who are preparing for the ISTQB Certified Tester-Foundation level exam based on the 2018 syllabus should try to go through the entire syllabus and then try to attempt practice exam in one sitting in 60 mins of time. Another approach is to read the syllabus chapter by chapter and then try to answer the questions from the respective chapters. The distribution of questions is as below:

Chapter 1 – Question 1-8

Chapter 2 – Question 9-13

Chapter 3 – Question 14-18

Chapter 4 – Question 19-29

Chapter 5 – Question 30-38

Chapter 6 – Question 39-40

Trademarks

The following registered trademarks and service marks are used in this document: ISTQB® is a registered trademark of the International Software Testing Qualifications Board.

2

Practice Exam 1

Question 1

Which of the following statements correctly relates to one of the software testing principles?

- a) The defects discovered during testing are mostly contained evenly in all the modules.
- b) If tests are not modified and used over and over again, they can no longer find any new defects.
- c) Testing all specified requirements will ensure that the system meets user expectations.
- d) Testing for e-commerce applications and social media applications are done similarly.

Select one option.

Question 2

Which of the following TWO techniques can be effective to the overall success of the project from a testing perspective?

- a) Involving testers to understand design with system designers while the system is being designed.
- b) Involving testers to create component level test cases for the developers before they start testing.
- c) Involving testers in the requirements reviews early in the project.
- d) Involving testers to run acceptance tests before the system users start their testing.

Select two options.

Question 3

Which of the following would **NOT** be considered as a valid objective for testing?

- a) To assess the quality of the system.
- b) To confirm the system works as expected.
- c) To release the system to customers on the planned date.
- d) To provide sufficient information to stakeholders regarding the quality of the test object.

Select one option.

Question 4

Differentiate the following test work products (1-4) by mapping them to the right categories (A-D).

- 1. Test procedures.
 - 2. Test charters.
 - 3. Test cases.
 - 4. Test progress reports.
-
- A. Test design work products.
 - B. Test monitoring and control work products.
 - C. Test analysis work products.
 - D. Test implementation work products.

- a) 1D, 2C, 3A, 4B.
- b) 1D, 2B, 3A, 4C.
- c) 1B, 2A, 3D, 4C.
- d) 1A, 2C, 3D, 4B.

Select one option.

Question 5

Which one of the following is the BEST description for test oracle?

- a) Work products produced during the test process for use in planning, designing, executing, evaluating and reporting.
- b) The degree to which a relationship can be established between two or more work products.
- c) A source to determine expected results to compare with the actual result of the system under test.
- d) The body of knowledge used as the basis for test analysis and design.

Select one option.

Question 6

In which stage of the ISTQB Test process the following three activities are performed?

- Review of the test basis.
 - Prioritization of the test conditions.
 - Creating bi-directional traceability between the test basis and the test conditions.
- a) Test implementation
 - b) Test execution
 - c) Test design
 - d) Test analysis

Select one option.

Question 7

Which of the following BEST describes the advantages of root cause analysis of defects?

- a) It can help in preparation of an action plan to effectively deal with failures.
- b) It can help in the early detection of defects.
- c) It can help in reduction of similar defects in the future.
- d) It can help in detecting false positive in the test results.

Select one option.

Question 8

Which of the following statements are TRUE about quality assurance?

- I. Quality assurance and testing both refer to the same set of activities.
- II. Quality assurance includes test activities that support the achievement of appropriate levels of quality.
- III. Quality assurance focuses on adherence to organizational processes.
- IV. Quality assurance focuses on root cause analysis and removal of defects.

- a) I and II
- b) III and IV
- c) I and III
- d) All of the above

Select one option.

Question 9

Which TWO of the following can be a trigger for maintenance testing?

- a) Decision to test that the system can work in conjunction with other systems.
- b) Decision to Introduce a new hardware device for an Internet of Things system.
- c) Decision to test the non-functional attributes for a new system.
- d) Decision to apply the patch to the system to fix the vulnerability issues.

Select two options.

Question 10

Which of the following statement BEST describes the V-model?

- a) Groups of features are specified, designed, build and tested together in a series of cycles.
- b) All of the testing activities start after the development phase is completed.
- c) There is a test level associated with each corresponding development phase.
- d) Establishing requirements, designing software and testing are done sequentially in increments.

Select one option.

Question 11

Which of the following TWO statements are TRUE with respect to regression testing and confirmation testing?

- a) Confirmation testing and regression testing can be both performed at all test levels.
- b) Confirmation tests find the root cause for defects and regression testing confirms that there are no side effects due to defect fixes.
- c) Confirmation testing can be performed only at the system level but regression testing can be performed at all test levels.
- d) Confirmation testing confirms that the original defect is fixed and regression testing confirms that that there are no side effects due to defect fixes.

Select two options.

Question 12

Match the different test level (1-4) with their respective objective (A-D) below?

1. Component testing.
 2. Integration testing.
 3. System testing.
 4. Acceptance testing.
-
- A. To verify the functional and non-functional behaviors of the system.
 - B. To gain user confidence in the system.
 - C. To find defects in the interfaces of the system.
 - D. To find defects in the individual modules of the system.
-
- a) 1D, 2B, 3A, 4C.
 - b) 1D, 2C, 3A, 4B.
 - c) 1D, 2C, 3B, 4A.
 - d) 1B, 2C, 3A, 4D.

Select one option.

Question 13

How can non-functional testing be applied during system testing?

- a) By verifying the robustness of the system in case of external call failure.
- b) By verifying that the presentation layer works on all supported browsers and mobile devices.
- c) By verifying that different group of people are able to use the system with ease.
- d) By verifying that data transferred between components are secure.

Select one option.

Question 14

The design workflow of a new loan system for a finance company is being reviewed. This system allows finance brokers to submit finance application on behalf of customers.

Once submitted the application goes to the loan approval-user work-flow for eligibility check. If the eligibility check is successful, it goes to the settlement-user work-flow for final approval.

In case of both approval and rejection of application a notification goes back to the broker work-flow.

The team is considering using different review techniques to verify the design workflow. Which of the following review technique will be MOST suitable for this situation?

- a) Scenario-based review technique.
- b) Checklist-based review technique.
- c) Role-based review technique
- d) Ad hoc review technique

Select one option.

Question 15

Pick the right sequence of the following activities for a work product review process:

- (A) Issue communication and analysis
- (B) Individual review
- (C) Planning
- (D) Fixing and reporting
- (E) Initiate review

- a) (C)→ (E)→ (B)→ (D)→ (A)
- b) (C)→ (B)→ (E)→ (A)→ (D)
- c) (C)→ (E)→ (B)→ (A)→ (D)
- d) (A)→ (C)→ (E)→ (B)→ (D)

Select one option.

Question 16

Match the following roles with their corresponding responsibilities for a formal review:

Roles:

1. Management
2. Author
3. Facilitator
4. Review leader

Responsibilities:

- A. Takes overall responsibility of the review.
 - B. Executes control decisions in the event of inadequate outcomes.
 - C. Fixes defects found during review.
 - D. Makes sure that the review meetings are successfully run.
-
- a) 1B, 2C, 3D, 4A.
 - b) 1D, 2C, 3B, 4A.
 - c) 1D, 2C, 3A, 4B.
 - d) 1C, 2A, 3D, 4B.

Select one option.

Question 17

Which of the below review types is the BEST option to choose when the review must follow specified entry and exit criteria?

- a) Walkthrough.
- b) Technical Review.
- c) Informal Review.
- d) Inspection.

Select one option.

Question 18

Which of the following defects are easier to find and fix through static testing?

- I. Unreachable code
- II. Deviations from coding standards
- III. Security vulnerability
- IV. Undefined variables

- a) I, II & III
- b) II, III & IV
- c) I, II & IV
- d) I, II, III & IV

Select one option.

Question 19

Which of the following explanations about statement testing is TRUE?

- I. Statement testing is most commonly used at the system test level.
- II. Statement testing exercises all the potential executable statements in the code more than once.
- III. Statement testing does not ensure that all the decision logic in the code are tested.
- IV. Statement testing is a white-box testing technique.

- a) I and II.
- b) III and IV.
- c) II and IV.
- d) II, III and IV.

Select one option.

Question 20

What is error-guessing test technique?

- a) A test technique where the tester dynamically designs and executes tests based on their knowledge, exploration of the test item and the results of previous tests.
- b) A test technique where experienced tester uses a set of rules or criteria for testing the software.
- c) A test technique where test cases are derived based on an analysis of the specification.
- d) A test technique in which tests are derived based on the tester's knowledge of past defects & failures.

Select one option.

Question 21

An electricity company charges a late fee to the customer if the payment is done after the bill due date.

The late fee is calculated based on the following rules:

- For outstanding amount \$500 or less the late fee is \$5
- For outstanding amount over \$500 and up to \$1000, the late fee is \$10
- For any outstanding amount more than \$1000 the late fee is \$15

How many tests will be required to test all set of values (\$) for three-point boundary value analysis? The outstanding amounts can only be in the nearest \$1.

- a) 10
- b) 9
- c) 8
- d) 7

Select one option.

Question 22

Which of the following TWO statements describing statement and decision testing is TRUE?

- a) In all situations, 100% statement coverage guarantees 100% decision coverage.
- b) Statement testing provides less coverage than decision testing.
- c) Statement testing finds defects in code that was not exercised by other tests while the decision coverage finds defects in the code where other tests have not taken both true and false outcomes.
- d) If there is no executable statement for the false outcome of decision than statement testing can provide the same coverage as decision testing.
- e) For 100% decision coverage, decision testing should be performed on the code only after the statement testing is completed.

Select two options.

Question 23

A bank offers new credit card customers certain discounts and extended interest-free periods for their first month of purchase.

Following decision table is used to calculate the discounts and extended interest-free period:

	Rule 1	Rule 2	Rule 3	Rule 4
Conditions				
First month credit card bill > \$5000	Y	Y	Y	N
Home loan account with the bank	N	Y	Y	N/A
Credit rating AAA	N	N	Y	N/A
Actions				
Discount	2%	3%	5%	0%
Extended Interest-free period	5 days	10 days	15 days	0 days

How much will be the discount and extended interest-free period for the following customer?

- John has a AAA credit rating and a home loan account with bank, his credit card bill for the first month is \$5000.
 - Rocky has a BBB credit rating and a home loan account with bank, his credit card bill for the first month is \$6000.
- a) John 5% discount, 15 days interest-free period; Rocky 2% discount, 5 interest-free period days.
 - b) John 5% discount, 15 days interest-free period; Rocky 3% discount, 10 interest-free period days.
 - c) John 0% discount, 0 days interest-free period; Rocky 3% discount, 5 interest-free period days.
 - d) John 0% discount, 0 days interest-free period, Rocky 3% discount, 10 interest-free period days.

Select one option.

Question 24

Which of the following factors can affect the selection of the test technique for a given system?

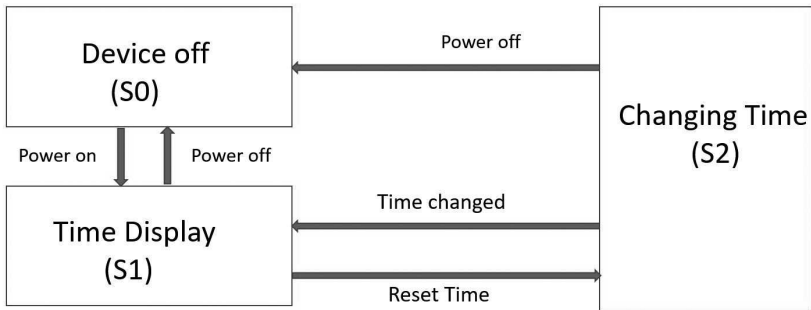
- I. Types of defects expected in the system.
- II. Customer or contractual requirements.
- III. Test data used for testing.
- IV. Tester knowledge and skills.

- a) I, II, III
- b) II, III, IV
- c) I, II, IV
- d) I, III, IV

Select one option.

Question 25

Consider the following state transition diagram showing the different states and input requests for a digital clock system.



How many test cases are required to test all the valid and invalid transactions in the above state transition diagram?

- Test cases for valid transactions = 5, Test cases for invalid transactions = 7.
- Test cases for valid transactions = 5, Test cases for invalid transactions = 5.
- Test cases for valid transactions = 5, Test cases for invalid transactions = 4.
- Test cases for valid transactions = 7, Test cases for invalid transactions = 7.

Select one option.

Question 26

Consider the following statement:

“The code subjected to test contains only one decision, and for both true and false outcomes there are executable statements.”

Which of the following explanation is correct with respect to the above scenario?

- a) For this scenario, the tests required for 100% decision coverage will be the same as the tests required for 100% statement coverage.
- b) For this scenario, the tests required for 100% decision coverage will be less than the tests required for 100% statement coverage.
- c) For this scenario, the tests required for 100% decision coverage will be more than the tests required for 100% statement coverage.
- d) With this information, it is not possible to get the exact number of tests to achieve 100% statement and 100% decision coverage.

Select one option.

Question 27

The monthly commission of the sales team is calculated based on the following set of rules:

- 0.5% on sales up to \$40,000
- 1.5% on sales over \$40,000 and up to \$100,000
- 2.5% on sales over \$100,000 and up to \$200,000
- 3% on sales over \$200,000

Which of the following groups of numbers (in \$) fall into three DIFFERENT equivalence classes?

- a) (\$200,000), (\$210,000), (\$240,000)
- b) (\$100,001), (\$140,000), (\$300,000)
- c) (\$40,000), (\$100,000), (\$200,000)
- d) (\$99,999), (\$100,000), (\$200,000)

Select one option.

Question 28

For which of the following situations in a project, exploratory testing will be the MOST suitable test technique?

1. Business process diagrams are available to describe user interactions with software.
 2. The specification documents are incomplete.
 3. A detailed list of possible defects and failures is available.
 4. There is a time pressure from business to complete testing in a reduced time frame.
-
- a) 1 & 2
 - b) 2 & 4
 - c) 2 & 3
 - d) 1 & 4

Select one option.

Question 29

The bank has a new one-year term deposit scheme where the interest rate is calculated based on the deposit amount.

- For deposit \$1000 or less, the interest rate is 1.5%.
- For deposit over \$1000 and up to \$10,000, the interest rate is 1.6%.
- For deposit over \$10,000 and up to \$25,000, the interest rate is 1.75%.
- For deposit more than \$25,000, the interest rate is 2%.

Which of the following is the most likely set of values (in \$) identified by two-point boundary value analysis? All the deposit amounts are in multiples of \$1?

- a) (1000), (1001), (10,000), (10,001), (25,000), (25,001)
- b) (1000), (10,000), (25,000)
- c) (0), (999), (1000), (10,000), (10,001), (25,000), (25001)
- d) (999), (1000), (1001), (10,000), (10,001), (25,000)

Select one option.

Question 30

Which of the following TWO activities are part of test planning?

- a) Evaluating the test basis.
- b) Test execution scheduling.
- c) Checking progress against schedule.
- d) Agreement on the structure for test documentation.
- e) Identifying test data.

Select two options.

Question 31

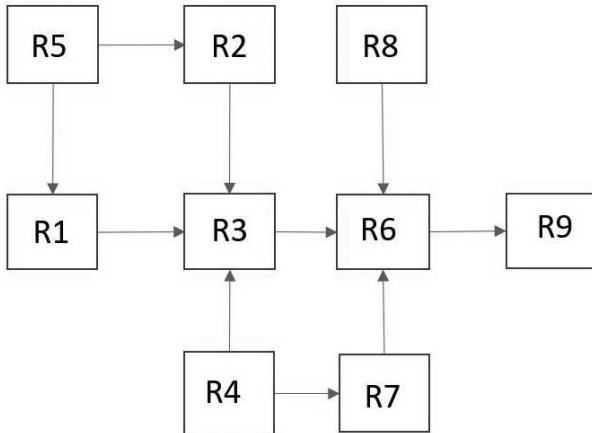
Which of the following are typical entry criteria for testing?

- a) Test tool availability, Test data availability, and security measures.
- b) Availability of testable code, time to market, and availability of test tools.
- c) Testable requirements availability, test environment availability, and test data availability.
- d) Test resources availability, Availability of testable code, performance measures.

Select one option.

Question 32

The following diagram shows the logical dependencies between a set of nine requirements, where a dependency is shown by an arrow. For example, “R3 → R6” means that R6 depends on R3.



Which one of the following options structures the test execution schedule according to the requirement dependencies?

- a) R1 → R3 → R5 → R2 → R4 → R7 → R6 → R8 → R9.
- b) R1 → R3 → R5 → R2 → R4 → R3 → R8 → R6 → R4 → R7 → R6 → R9.
- c) R5 → R2 → R5 → R1 → R3 → R4 → R3 → R7 → R6 → R8 → R9.
- d) R5 → R1 → R2 → R4 → R3 → R7 → R8 → R6 → R9.

Select one option.

Question 33

Which of the following BEST describes the task carried out by the testers in the project?

- a) Introduce suitable metrics for measuring test progress.
- b) Determine the test schedule.
- c) Prepare and acquire test data.
- d) Prepare and deliver test progress report to the stakeholders.

Select one option.

Question 34

Which of the following metrics can be MOST useful to assess the current quality of the test object?

- a) Details of test cases run/not run.
- b) Percentage of work done in test environment preparation.
- c) Details of defects found and fixed during test execution.
- d) Cost of testing.

Select one option.

Question 35

Which of the following BEST describes the difference between the test progress report and test summary report?

- a) The test progress report is prepared for the internal stakeholders, whereas a test summary report is prepared for external stakeholders.
- b) The test progress report covers only the status of test activities, whereas a test summary report covers only the quality of the test object.
- c) The test progress report is prepared at regular intervals after the entry criteria are reached, whereas a test summary report is prepared after the exit criteria is reached for testing.
- d) The test progress report is prepared at the end of each test activity, whereas a test summary report is prepared during the test activity.

Select one option.

Question 36

Which of the following is NOT a characteristic of the metrics-based test estimation technique?

- a) Project budget from similar projects in the past can be used for estimation.
- b) Defects data from similar projects in the past can be used for estimation.
- c) Average of estimates collected from technical experts can be used for estimations.
- d) Standard timeframe to set up a test environment from similar projects can be used for estimation.

Select one option.

Question 37

The project for the enhancement of a banking application is in the last phase of testing. During the regression testing, the tester has found that some of the customers are not able to generate the monthly account statement from the website. This functionality works fine from the mobile app for the same users.

Tester has written the defect report to highlight this issue with the following information:

Defect ID: 139

Title: Monthly account statement functionality not working

Summary: After the user clicks on the statement button on the account page, monthly account statement is not generated. The same users are able to generate the monthly account statement from the mobile app.

Severity of defect: High

Priority of defect: High

Expected result: All the users should be able to generate the monthly account statement.

Actual result: Some of the users are not able to generate the monthly account statement.

What is the MOST valuable information that is missed by the tester in the above defect report?

- a) Test case reference for this defect.
- b) Test environment and detected phase.
- c) Date and author of the defect report.
- d) Data used for testing.

Select one option.

Question 38

Which of the following statement describing the different type of test strategies is false?

- a) Model-based test strategy involves designing tests based on some model for a required aspect of the product.
- b) Consultative test strategy involves designing tests based on the advice, guidance, or instructions of stakeholders and business domain experts.
- c) Analytical test strategy involves designing tests based on industry-specific standards, or by processes imposed by the organization.
- d) Methodical test strategy involves designing tests based on some predefined set of test conditions or likely types of failures.

Select one option.

Question 39

Which of the following cannot be considered as a valid objective of a pilot project while introducing a new tool into an organization?

- a) Assess whether the benefits will be achieved at a reasonable cost.
- b) Understand the strength and weakness of the tool.
- c) To identify the tool owner who can implement the tool in the organization.
- d) Evaluate how the tool fits in with the existing processes and practices.

Select one option.

Question 40

Which of the following tool is most useful for the early detection of integration problems and conflicting changes?

- a) Performance testing tool
- b) Continuous integration tool
- c) Coverage tool
- d) Static analysis tool

Select one option.

Answers

1	B	K2
2	A, C	K2
3	C	K1
4	A	K2
5	C	K1
6	D	K2
7	C	K2
8	B	K2
9	B, D	K2
10	C	K2
11	A, D	K2
12	B	K2
13	B	K1
14	C	K3
15	C	K2
16	A	K1
17	D	K2
18	D	K2
19	B	K2
20	D	K1

21	C	K3
22	B, C	K2
23	D	K3
24	C	K2
25	A	K3
26	A	K2
27	C	K3
28	B	K2
29	A	K3
30	B, D	K2
31	C	K2
32	D	K3
33	C	K1
34	C	K1
35	C	K2
36	C	K2
37	D	K3
38	C	K2
39	C	K1
40	B	K2

3

Answers – Practice Exam 1

Question 1

Which of the following statements correctly relates to one of the software testing principles?

- a) The defects discovered during testing are mostly contained evenly in all the modules.
- b) **If tests are not modified and used over and over again, they can no longer find any new defects.**
- c) Testing all specified requirements will ensure that the system meets user expectations.
- d) Testing for e-commerce applications and social media applications are done similarly.

FL-1.3.1 (K2) Explain the seven testing principles

Justification

- a) Not Correct – Defects are not evenly distributed but clustered together (**Principle#4**).
- b) **Correct** – If the same tests are not modified and repeated over and over again, eventually these tests will no longer find any new defects (**Principle#5**).
- c) Not Correct – Testing all specified requirements can't guarantee that the system is fulfilling user needs (**Principle#7**).
- d) Not Correct– Testing for e-commerce application and social media application are done differently as testing is context dependent (**Principle#6**).

Question 2

Which of the following TWO techniques can be effective to the overall success of the project from a testing perspective?

- a) **Involving testers to understand design with system designers while the system is being designed.**
- b) Involving testers to create component level test cases for the developers before they start testing.
- c) **Involving testers in the requirements reviews early in the project.**
- d) Involving testers to run acceptance tests before the system users start their testing.

FL-1.2.1 (K2) Give examples of why testing is necessary**Justification**

- a) **Correct** – Involving testers with system designers while the system is being designed can increase each party's understanding of the design and how to test it. This increased understanding can reduce the risk of fundamental design defects and enable tests to be identified at an early stage. (Syllabus 1.2.1)
- b) Not correct – Component test case creation requires a thorough knowledge of the component. Therefore, developers are best to create and execute component level test cases.
- c) **Correct**– Having testers involved in requirements reviews could detect defects in these work products early. The identification and removal of requirements defects reduce the risk of incorrect or untestable features being developed. (Syllabus 1.2.1)
- d) Not correct – Acceptance testing requires running the test cases from user's perspective so it is better if system users can run the acceptance tests. Moreover, the re-execution of tests by testers will increase testing time and effort.

Question 3

Which of the following would **NOT** be considered as a valid objective for testing?

- a) To assess the quality of the system.
- b) To confirm the system works as expected.
- c) **To release the system to customers on the planned date.**
- d) To provide sufficient information to stakeholders regarding the quality of the test object.

FL-1.1.1 (K1) Identify typical objectives of testing**Justification**

- a) Not correct – This is a valid objective for testing. (Syllabus 1.1.1)
- b) Not correct – This is a valid objective for testing. (Syllabus 1.1.1)
- c) **Correct** – This is not a valid objective for testing. Testing cannot guarantee that the System will be released to customers on the planned date.
- d) Not correct – This is a valid objective for testing (Syllabus 1.1.1)

Question 4

Differentiate the following test work products (1-4) by mapping them to the right categories (A-D).

1. Test procedures.
 2. Test charters.
 3. Test cases.
 4. Test progress reports.
-
- A. Test design work products.
 - B. Test monitoring and control work products.
 - C. Test analysis work products.
 - D. Test implementation work products.

- a) **1D, 2C, 3A, 4B.**
- b) 1D, 2B, 3A, 4C.
- c) 1B, 2A, 3D, 4C.
- d) 1A, 2C, 3D, 4B.

FL-1.4.3 (K2) Differentiate the work products that support the test process

Justification

- **Test procedures** are Test implementation work products. Syllabus 1.4.3. **(1D)**
- **Test charters** are Test analysis work products. Syllabus 1.4.3. **(2C)**
- **Test cases** are Test design work products. Syllabus 1.4.3. **(3A)**
- **Test progress reports** are Test monitoring and control work products. Syllabus 1.4.3. **(4B)**

Thus:

- a) **Correct**
- b) Not correct
- c) Not correct
- d) Not correct

Question 5

Which one of the following is the BEST description for test oracle?

- a) Work products produced during the test process for use in planning, designing, executing, evaluating and reporting.
- b) The degree to which a relationship can be established between two or more work products.
- c) **A source to determine expected results to compare with the actual result of the system under test.**
- d) The body of knowledge used as the basis for test analysis and design.

FL-1.x (K1) Keywords Chapter 1

Justification

- a) Not correct – This is the definition of testware according to Glossary.
- b) Not correct – This is the definition of traceability according to Glossary.
- c) **Correct** – This is the definition of test oracle according to Glossary.
- d) Not correct – This is the definition of test basis according to Glossary.

Question 6

In which stage of the ISTQB Test process the following three activities are performed?

- Review of the test basis.
 - Prioritization of the test conditions.
 - Creating bi-directional traceability between the test basis and the test conditions.
- a) Test implementation
 - b) Test execution
 - c) Test design
 - d) **Test analysis**

FL-1.4.2 (K2) Describe the test activities and respective tasks within the test process

Justification

- a) Not correct – these activities are not part of Test implementation.
- b) Not correct – these activities are not part of Test execution.
- c) Not correct – these activities are not part of Test design.
- d) **Correct** – these activities are performed during the Test analysis. Syllabus 1.4.2.

Question 7

Which of the following BEST describes the advantages of root cause analysis of defects?

- a) It can help in preparation of an action plan to effectively deal with failures.
- b) It can help in the early detection of defects.
- c) **It can help in reduction of similar defects in the future.**
- d) It can help in detecting false positive in the test results.

FL-1.2.4 (K2) Distinguish between the root cause of a defect and its effects

Justification

- a) Not correct- This is not the function of root cause analysis of defects.
- b) Not correct - This is not the function of root cause analysis of defects.
- c) **Correct** - Root cause of defects are the earliest actions or conditions that contributed to creating the defects. Root cause analysis can lead to process improvements that prevent a significant number of future defects. (Syllabus 1.2.4)
- d) Not correct - This is not the function of root cause analysis of defects.

Question 8

Which of the following statements are TRUE about quality assurance?

- I. Quality assurance and testing both refer to the same set of activities.
 - II. Quality assurance includes test activities that support the achievement of appropriate levels of quality.
 - III. Quality assurance focuses on adherence to organizational processes.
 - IV. Quality assurance focuses on root cause analysis and removal of defects.
- a) I and II
 - b) III and IV**
 - c) I and III
 - d) All of the above

Select one option.

FL-1.2.2 (K2) Describe the relationship between testing and quality assurance and give examples of how testing contributes to higher quality

Justification

- I. This is **false**. Quality assurance and testing are not the same but are related. Syllabus 1.2.2
- II. This is **false**. It is not quality assurance but quality control which including test activities, that support the achievement of appropriate levels of quality. Syllabus 1.2.2
- III. This is **true**. Quality assurance is typically focused on adherence to proper processes, in order to provide confidence that the appropriate levels of quality will be achieved. Syllabus 1.2.2
- IV. This is **true**. Quality assurance focuses on root cause analysis to detect and remove the causes of defects. Syllabus 1.2.2

Thus:

- a) Not correct
- b) Correct**
- c) Not correct
- d) Not correct

Question 9

Which TWO of the following can be a trigger for maintenance testing?

- a) Decision to test that the system can work in conjunction with other systems.
- b) **Decision to Introduce a new hardware device for an Internet of Things system.**
- c) Decision to test the non-functional attributes for a new system.
- d) **Decision to apply the patch to the system to fix the vulnerability issues.**

FL-2.4.1 (K2) Summarize triggers for maintenance testing

Justification

- a) Not correct – this is a trigger for integration testing, not maintenance testing.
- b) **Correct** – For Internet of Things (IoT) system maintenance testing may be triggered by the introduction of a new or modified hardware device and software services (Syllabus 2.4.1).
- c) Not correct – this is part of system testing, not maintenance testing.
- d) **Correct** – Patches to fix defects and vulnerability issues require maintenance testing (Syllabus 2.4.1).

Question 10

Which of the following statement BEST describes the V-model?

- a) Groups of features are specified, designed, build and tested together in a series of cycles.
- b) All of the testing activities start after the development phase is completed.
- c) **There is a test level associated with each corresponding development phase.**
- d) Establishing requirements, designing software and testing are done sequentially in increments.

FL-2.1.1 (K2) Explain the relationship between software development activities and test activities in the software life cycle

Justification

- a) Not correct – This describes the iterative model.
- b) Not correct – This describes the waterfall model.
- c) **Correct** – This describes the V-model. (Syllabus 2.1.1)
- d) Not correct – This describes the incremental development.

Question 11

Which of the following TWO statements are TRUE with respect to regression testing and confirmation testing?

- a) **Confirmation testing and regression testing can be both performed at all test levels.**
- b) Confirmation tests find the root cause for defects and regression testing confirms that there are no side effects due to defect fixes.
- c) Confirmation testing can be performed only at the system level but regression testing can be performed at all test levels.
- d) **Confirmation testing confirms that the original defect is fixed and regression testing confirms that there are no side effects due to defect fixes.**

FL-2.3.3 (K2) Compare the purposes of confirmation testing and regression testing

Justification

- a) **Correct** – This is true. Confirmation testing and regression testing are performed at all test levels. (Syllabus 2.3.4)
- b) Not correct– This is false. Confirmation testing does not find the root cause for defects.
- c) Not correct – This is false. Confirmation testing can be performed at all test levels.
- d) **Correct** – This is true. Confirmation testing confirms that the original defect is fixed and regression testing confirms that there are no side effects due to defect fixes. (Syllabus 2.3.4)

Question 12

Match the different test level (1-4) with their respective objective (A-D) below?

1. Component testing.
 2. Integration testing.
 3. System testing.
 4. Acceptance testing.
-
- A. To verify the functional and non-functional behaviors of the system.
 - B. To gain user confidence in the system.
 - C. To find defects in the interfaces of the system.
 - D. To find defects in the individual modules of the system.
-
- a) 1D, 2B, 3A, 4C.
 - b) 1D, 2C, 3A, 4B.**
 - c) 1D, 2C, 3B, 4A.
 - d) 1B, 2C, 3A, 4D.

FL-2.2.1 (K2) Compare the different test levels from the perspective of objectives, test basis, test objects, typical defects and failures, and approaches and responsibilities

Justification

- The objective of **component testing** is to find defects in the individual modules of the system. Syllabus 2.2.1. **(1D)**
- The objective of **Integration testing** finds defects in the system interfaces. Syllabus 2.2.2. **(2C)**
- The objective of **system testing** is to verify the functional and non-functional behaviors of the system. Syllabus 2.2.3. **(3A)**
- The objective of **acceptance testing** is to gain user confidence in the system. Syllabus 2.2.4. **(4B)**

Thus:

- a) Not correct
- b) Correct**
- c) Not correct
- d) Not correct

Question 13

How can non-functional testing be applied during system testing?

- a) By verifying the robustness of the system in case of external call failure.
- b) **By verifying that the presentation layer works on all supported browsers and mobile devices.**
- c) By verifying that different group of people are able to use the system with ease.
- d) By verifying that data transferred between components are secure.

FL-2.3.2 (K1) Recognize that functional, non-functional and white-box tests occur at any test level

Justification

- a) Not correct – This is relevant for system integration testing where **reliability** tests are designed to evaluate the system robustness in case of external call failure.
- b) **Correct** –For system testing, **portability** tests are designed to check whether the presentation layer works on all supported browsers and mobile devices. (Syllabus 2.3.5)
- c) Not correct – This is relevant for acceptance testing where **usability** tests are designed to evaluate the accessibility of the customer interface for a different group of people e.g. people with disability.
- d) Not correct – This is relevant for component integration testing where **security** tests are designed for buffer overflow vulnerabilities for data transferred between components.

Question 14

The design workflow of a new loan system for a finance company is being reviewed. This system allows finance brokers to submit finance application on behalf of customers.

Once submitted the application goes to the loan approval-user workflow for eligibility check. If the eligibility check is successful, it goes to the settlement-user work-flow for final approval.

In case of both approval and rejection of application a notification goes back to the broker work-flow.

The team is considering using different review techniques to verify the design workflow. Which of the following review technique will be MOST suitable for this situation?

- a) Scenario-based review technique.
- b) Checklist-based review technique.
- c) **Role-based review technique**
- d) Ad hoc review technique

FL-3.2.4 (K3) Apply a review technique to a work product to find defects**Justification**

- a) Not correct – There is no reference to the different scenarios in this situation. Moreover, review of design is required from the perspective of the individual roles.
- b) Not correct – This is a new system and there are no details of typical defects which need to be covered therefore checklist-based review is not suitable for this situation.
- c) **Correct** – As mentioned in the question there are different user roles for this system (Broker, approval-user, settlement-user). Therefore role-based review technique will be most suitable to cover the design from the perspective of the end users.
- d) Not correct – This situation needs a structured approach to cover all the different roles of the system, ad hoc review is not suitable for this situation.

Question 15

Pick the right sequence of the following activities for a work product review process:

- (A) Issue communication and analysis
- (B) Individual review
- (C) Planning
- (D) Fixing and reporting
- (E) Initiate review

- a) (C)→ (E)→ (B)→ (D)→ (A)
- b) (C)→ (B)→ (E)→ (A)→ (D)
- c) **(C)→ (E)→ (B)→ (A)→ (D)**
- d) (A)→ (C)→ (E)→ (B)→ (D)

FL-3.2.1 (K2) Summarize the activities of the work product review process

Justification

- a) Not correct – See c) for the right sequence.
- b) Not correct – See c) for the right sequence.
- c) **Correct** – The right sequence of activities of formal review are:
Planning→ Initiate review→ Individual review→ Issue communication and analysis→ Fixing and reporting. (Syllabus 3.2.1)
- d) Not correct – See c) for the right sequence.

Question 16

Match the following roles with their corresponding responsibilities for a formal review:

Roles:

1. Management
2. Author
3. Facilitator
4. Review leader

Responsibilities:

- A. Takes overall responsibility of the review.
- B. Executes control decisions in the event of inadequate outcomes.
- C. Fixes defects found during review.
- D. Makes sure that the review meetings are successfully run.

- a) **1B, 2C, 3D, 4A.**
- b) 1D, 2C, 3B, 4A.
- c) 1D, 2C, 3A, 4B.
- d) 1C, 2A, 3D, 4B.

FL-3.2.2 (K1) Recognize the different roles and responsibilities in a formal review

Justification

1. Management executes control decisions in the event of inadequate outcomes. (Syllabus 3.2.2) **(1B)**
2. Author fixes defects in the work product which are found during review. (Syllabus 3.2.2) **(2C)**
3. Facilitator or moderator ensures that the review meetings are successfully run. (Syllabus 3.2.2) **(3D)**
4. Review leader takes overall responsibility for the review. (Syllabus 3.2.2) **(4A)**

Thus:

- a) **Correct**
- b) Not correct
- c) Not correct
- d) Not correct

Question 17

Which of the below review types is the BEST option to choose when the review must follow specified entry and exit criteria?

- a) Walkthrough.
- b) Technical Review.
- c) Informal Review.
- d) **Inspection.**

FL-3.2.3 (K2) Explain the differences between different review types: informal review, walkthrough, technical review and inspection

Justification

- a) Not correct – Walkthrough does not use a formal process such as entry and exit criteria.
- b) Not correct – For Technical Reviews entry and exit criteria is not must.
- c) Not correct – Informal review does not use entry and exit criteria.
- d) **Correct** – Inspection follows the most formal process with specified entry and exit criteria (Syllabus 3.2.3)

Question 18

Which of the following defects are easier to find and fix through static testing?

- I. Unreachable code
- II. Deviations from coding standards
- III. Security vulnerability
- IV. Undefined variables

- a) I, II & III
- b) II, III & IV
- c) I, II & IV
- d) **I, II, III & IV**

FL-3.1.3 (K2) Explain the difference between static and dynamic techniques, considering objectives, types of defects to be identified, and the role of these techniques within the software lifecycle

Justification

- I. Coding defects such as unreachable code are easier to find and fix through static testing. (Syllabus 3.1.3)
- II. Deviation from coding standards is easier to find and fix through static testing. (Syllabus 3.1.3)
- III. Security vulnerabilities such as susceptibility of buffer overflows are easier to find and fix through static testing. (Syllabus 3.1.3)
- IV. Coding defects such as undefined variables are easier to find and fix through static testing. (Syllabus 3.1.3)

Thus:

- a) Not correct
- b) Not correct
- c) Not correct
- d) **Correct**

Question 19

Which of the following explanations about statement testing is TRUE?

- I. Statement testing is most commonly used at the system test level.
- II. Statement testing exercises all the potential executable statements in the code more than once.
- III. Statement testing does not ensure that all the decision logic in the code are tested.
- IV. Statement testing is a white-box testing technique.

- a) I and II.
- b) III and IV.**
- c) II and IV.
- d) II, III and IV.

Select one option.

FL-4.3.1 (K2) Explain statement coverage

Justification

- I. This is false, statement testing is most commonly used at the **component test** level. (Syllabus 4.3)
- II. This is false, statement testing exercises all the potential executable statements in the code at **least once**. (Syllabus 4.3.3)
- III. This is true, statement testing does not ensure that all the decision logic in the code are tested. (Syllabus 4.3.3)
- IV. This is true, statement testing is a white-box testing technique. (Syllabus 4.3)

Thus:

- a) Not correct
- b) Correct**
- c) Not correct
- d) Not correct

Question 20

What is error-guessing test technique?

- a) A test technique where the tester dynamically designs and executes tests based on their knowledge, exploration of the test item and the results of previous tests.
- b) A test technique where experienced tester uses a set of rules or criteria for testing the software.
- c) A test technique where test cases are derived based on an analysis of the specification.
- d) **A test technique in which tests are derived based on the tester's knowledge of past defects & failures.**

FL-4.x (K1) Keywords**Justification**

- a) Not correct – This is the definition of exploratory testing from the Glossary.
- b) Not correct – This is the definition of checklist-based testing from the Glossary.
- c) Not correct – This is the definition of black-box test technique from the Glossary.
- d) **Correct** – This is the definition of error-guessing testing from the Glossary.

Question 21

An electricity company charges a late fee to the customer if the payment is done after the bill due date.

The late fee is calculated based on the following rules:

- For outstanding amount \$500 or less the late fee is \$5
- For outstanding amount over \$500 and up to \$1000, the late fee is \$10
- For any outstanding amount more than \$1000 the late fee is \$15

How many tests will be required to test all set of values (\$) for three-point boundary value analysis? The outstanding amounts can only be in the nearest \$1.

- a) 10
- b) 9
- c) 8
- d) 7

FL-4.2.2 (K3) Apply boundary value analysis to derive test cases from given requirements

Justification

The following partitions can be identified:

- | | | |
|----|------------|---|
| 1. | – 500 | Three-point boundaries 499,500,501 |
| 2. | 501 – 1000 | Three -point boundaries 500, 501, 502, 999, 1000,1001 |
| 3. | 1001 – | Three -point boundaries 1000,1001,1002 |

Test are required for boundaries 499, 500, 501, 502, 999, 1000, 1001, 1002

Therefore, total test required to test three-point boundaries is 8.

Thus:

- a) Not correct – see the justification above.
- b) Not correct – see the justification above.
- c) **Correct** – Includes all three-point boundary values.
- d) Not correct – see the justification above.

Question 22

Which of the following TWO statements describing statement and decision testing is TRUE?

- a) In all situations, 100% statement coverage guarantees 100% decision coverage.
- b) **Statement testing provides less coverage than decision testing.**
- c) **Statement testing finds defects in code that was not exercised by other tests while the decision coverage finds defects in the code where other tests have not taken both true and false outcomes.**
- d) If there is no executable statement for the false outcome of decision than statement testing can provide the same coverage as decision testing.
- e) For 100% decision coverage, decision testing should be performed on the code only after the statement testing is completed.

FL-4.3.3 (K2) Explain the value of statement and decision coverage

Justification

- a) Not correct – This statement is false. Achieving 100% decision coverage guarantees 100% statement coverage (but not vice versa). (Syllabus 4.3.3)
- b) **Correct** – The statement is true. As the decisions are not tested in statement testing it provides less coverage than decision testing. (Syllabus 4.3.3)
- c) **Correct** – This statement is true. (Syllabus 4.3.3)
- d) Not correct – This statement is false. Even though there is no executable statement in the false outcome of the decision, testing is required to see the results when the decision is false. Therefore, statement testing can't provide the same coverage.
- e) Not correct – This statement is false. Both decision and statement testing are independent of each other.

Question 23

A bank offers new credit card customers certain discounts and extended interest-free periods for their first month of purchase.

Following decision table is used to calculate the discounts and extended interest-free period:

	Rule 1	Rule 2	Rule 3	Rule 4
Conditions				
First month credit card bill > \$5000	Y	Y	Y	N
Home loan account with the bank	N	Y	Y	N/A
Credit rating AAA	N	N	Y	N/A
Actions				
Discount	2%	3%	5%	0%
Extended Interest-free period	5 days	10 days	15 days	0 days

How much will be the discount and extended interest-free period for the following customer?

- John has a AAA credit rating and a home loan account with bank, his credit card bill for the first month is \$5000.
 - Rocky has a BBB credit rating and a home loan account with bank, his credit card bill for the first month is \$6000.
- a) John 5% discount, 15 days interest-free period; Rocky 2% discount, 5 interest-free period days.
 - b) John 5% discount, 15 days interest-free period; Rocky 3% discount, 10 interest-free period days.
 - c) John 0% discount, 0 days interest-free period; Rocky 3% discount, 5 interest-free period days.
 - d) **John 0% discount, 0 days interest-free period, Rocky 3% discount, 10 interest-free period days.**

FL-4.2.3 (K3) Apply decision table testing to derive test cases from given requirements

Justification

John falls under **Rule 4** hence 0% discount and 0 days interest-free period.

Rocky falls under **Rule 2** hence 3% discount and 10 days interest-free period.

Hence,

- a) Not correct- see explanation above.
- b) Not correct- see explanation above.
- c) Not correct- For John the discount and interest-free period are correct but for Rocky, it will 3% discount and 10 days interest-free period. See explanation above.
- d) **Correct** - see explanation above.

Question 24

Which of the following factors can affect the selection of the test technique for a given system?

- I. Types of defects expected in the system.
 - II. Customer or contractual requirements.
 - III. Test data used for testing.
 - IV. Tester knowledge and skills.
- a) I, II, III
 - b) II, III, IV
 - c) **I, II, IV**
 - d) I, III, IV

FL-4.1.1 (K2) Explain the characteristics, commonalities, and differences between black-box test techniques, white-box test techniques and experience-based test techniques

Justification

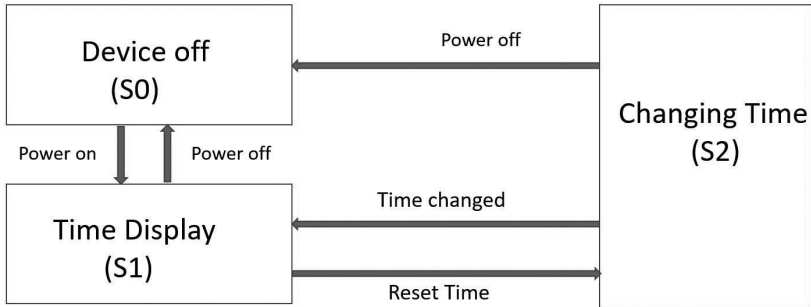
- I. This can affect the selection of test techniques for the system. (Syllabus 4.1.1)
- II. This can affect the selection of test techniques for the system. (Syllabus 4.1.1)
- III. This will not affect the selection. It is the purpose of the test technique to actually help in identification of test data. (Syllabus 4.1.1)
- IV. This can affect the selection of test techniques for the system. (Syllabus 4.1.1)

Thus:

- a) Not correct
- b) Not correct
- c) **Correct**
- d) Not correct

Question 25

Consider the following state transition diagram showing the different states and input requests for a digital clock system.



How many test cases are required to test all the valid and invalid transactions in the above state transition diagram?

- Test cases for valid transactions = 5, Test cases for invalid transactions = 7.**
- Test cases for valid transactions = 5, Test cases for invalid transactions = 5.
- Test cases for valid transactions = 5, Test cases for invalid transactions = 4.
- Test cases for valid transactions = 7, Test cases for invalid transactions = 7.

FL-4.2.4 (K3) Apply state transition testing to derive test cases from given requirements

Justification

A state table can be created for the above state transition diagram. All the states are on the left-hand side of the table, and all the triggers(events) are across the top.

Now to derive the value of each cell of the table input the trigger at the top of the current state.

Example: When the current state is **S0** and '**Power on**' trigger is applied the resulting state (end state) is **S1**. This is a valid transaction because when the trigger is applied the end state is different from the current state.

When the current state is **S1** and '**Power on**' trigger is applied nothing happens. Therefore, this an invalid transaction or “null” transaction (denoted by '-').

	Power on	Power off	Time changed	Reset Time
S0	S1	-	-	-
S1	-	S0	-	S2
S2	-	S0	S1	-

From the table, there are 5 valid transactions and 7 invalid or “null” transactions (denoted by '-').

Therefore, test cases for valid transactions =5, Test cases for valid transactions = 7

- Correct** – see the explanation above.
- Not correct – see the explanation above.
- Not correct – see the explanation above.
- Not correct – see the explanation above.

Question 26

Consider the following statement:

“The code subjected to test contains only one decision, and for both true and false outcomes there are executable statements.”

Which of the following explanation is correct with respect to the above scenario?

- a) **For this scenario, the tests required for 100% decision coverage will be the same as the tests required for 100% statement coverage.**
- b) For this scenario, the tests required for 100% decision coverage will be less than the tests required for 100% statement coverage.
- c) For this scenario, the tests required for 100% decision coverage will be more than the tests required for 100% statement coverage.
- d) With this information, it is not possible to get the exact number of tests to achieve 100% statement and 100% decision coverage.

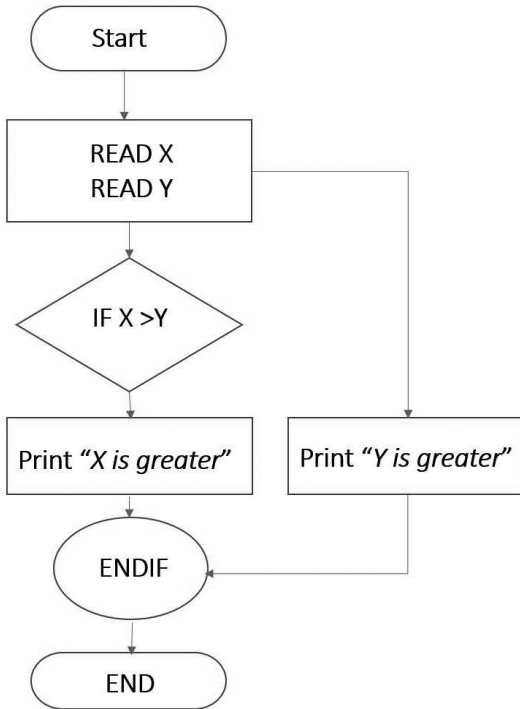
FL-4.3.2 (K2) Explain decision coverage

Justification

- a) **Correct** – For this scenario, the tests required for 100% decision coverage will be the same as the tests required for 100% statement coverage. See explanation below.
- b) Not correct – This is never possible, tests required for 100% decision coverage will be always equal or more than the tests required for 100% statement coverage.
- c) Not correct – See explanation below.
- d) Not correct – There is enough information, see explanation below.

Following code is have one decision with an executable statement for each outcome:

1. DEC X, Y integer
2. READ X, READ Y
3. IF X > Y THEN
4. “X is greater than Y”
5. ELSE
6. “Y is greater than X”
7. END IF

Flowchart for this code:

From the flow chart, it is clear that two tests are required to achieve 100% decision coverage and 100% statement coverage.

For the first test, the values are $X=100$, $Y=50$

For the second test, the values are $X=50$, $Y=100$

Thus, tests for 100% decision coverage will be the same as the tests required for 100% statement coverage.

Question 27

The monthly commission of the sales team is calculated based on the following set of rules:

- 0.5% on sales up to \$40,000
- 1.5% on sales over \$40,000 and up to \$100,000
- 2.5% on sales over \$100,000 and up to \$200,000
- 3% on sales over \$200,000

Which of the following groups of numbers (in \$) fall into three DIFFERENT equivalence classes?

- a) (\$200,000), (\$210,000), (\$240,000)
- b) (\$100,001), (\$140,000), (\$300,000)
- c) **(\$40,000), (\$100,000), (\$200,000)**
- d) (\$99,999), (\$100,000), (\$200,000)

FL-4.2.1 (K3) Apply equivalence partitioning to derive test cases from given requirements

Justification

- a) Not correct – (\$210,000), (\$240,000) are from same partitions. see explanation in c).
- b) Not correct – (\$100,001), (\$140,000) are from same partitions. see explanation in c).
- c) **Correct** – Partitions are as below:
 1. equivalence partition: $0 < \text{sales} \leq \$40,000$
 2. equivalence partition: $\$40,000 < \text{sales} \leq \$100,000$
 3. equivalence partition: $\$100,000 < \text{sales} \leq \$200,000$
 4. equivalence partition: $\$200,000 < \text{sales}$.
 \$40,000 is from 1st partition, \$100,000 from 2nd partition and \$200,000 from 3rd partition.
- d) Not correct – (\$99,999), (\$100,000) are from same partitions. see explanation in c).

Question 28

For which of the following situations in a project, exploratory testing will be the MOST suitable test technique?

1. Business process diagrams are available to describe user interactions with software.
 2. The specification documents are incomplete.
 3. A detailed list of possible defects and failures is available.
 4. There is a time pressure from business to complete testing in a reduced time frame.
-
- a) 1 & 2
 - b) 2 & 4**
 - c) 2 & 3
 - d) 1 & 4

FL-4.4.2 (K2) Explain exploratory testing**Justification**

1. When business process diagrams are available, use case testing is most suitable. (Syllabus 4.2.5)
2. When there are few inadequate specifications, **exploratory testing** is most suitable. (Syllabus 4.4.2)
3. When a details list of possible defects and failures is available, error guessing technique is most suitable. (Syllabus 4.4.1)
4. When there is significant time pressure on testing, **exploratory testing** is most suitable. (Syllabus 4.4.2)

Thus:

- a) Not correct
- b) Correct**
- c) Not correct
- d) Not correct

- For deposit \$1000 or less, the interest rate is 1.5%.
- For deposit over \$1000 and up to \$10,000, the interest rate is 1.6%.
- For deposit over \$10,000 and up to \$25,000, the interest rate is 1.75%.
- For deposit more than \$25,000, the interest rate is 2%.

a) (1000), (1001), (10,000), (10,001), (25,000), (25,001)
b) (1000), (10,000), (25,000)
c) (0), (999), (1000), (10,000), (10,001), (25,000), (25001)
d) (999), (1000), (1001), (10,000), (10,001), (25,000)

Justification

1.	– 1000	Two-point boundaries (1000), (1001)
2.	1001 – 10,000	Two-point boundaries (1000), (1001), (10,000), (10,001)
3.	10,001 – 25,000	Two-point boundaries (10,000), (10,001), (25,000), (25,001)
4.	25,001 –	Two-point boundaries (25,000), (25,001)

- a) **Correct** – Includes all two-point boundary values.
- b) Not correct – Does not include all two-point boundary values.
- c) Not correct – Does not include all two-point boundary values. There are also values not necessary for two-point boundary value analysis.
- d) Not correct – Does not include all two-point boundary values. There are also values not necessary for two-point boundary value analysis.

Question 30

Which of the following TWO activities are part of test planning?

- a) Evaluating the test basis.
- b) Test execution scheduling.**
- c) Checking progress against schedule.
- d) Agreement on the structure for test documentation.**
- e) Identifying test data.

FL-5.2.1 (K2) Summarize the purpose and content of a test plan

Justification

- a) Not correct –It is a part of test analysis.
- b) Correct** – Scheduling for all the test activities is an important part of planning. (Syllabus 5.2.1)
- c) Not correct – It is a part of test monitoring and control.
- d) Correct** – During planning it is necessary to agree on the structure and level of detail for test documentation. (Syllabus 5.2.1)
- e) Not correct – It is a part of test design.

Question 31

Which of the following are typical entry criteria for testing?

- a) Test tool availability, Test data availability, and security measures.
- b) Availability of testable code, time to market, and availability of test tools.
- c) Testable requirements availability, test environment availability, and test data availability.**
- d) Test resources availability, Availability of testable code, performance measures.

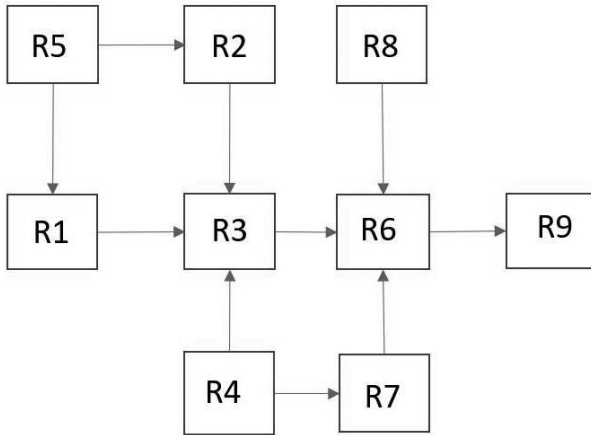
FL-5.2.3 (K2) Give examples of potential entry and exit criteria

Justification

- a) Not correct – “Security measures” is typically a part of the exit criteria.
- b) Not correct – “Time to market” is typically a part of the exit criteria
- c) Correct** – All of them are part of typical entry criteria for testing. (Syllabus 5.2.3)
- d) Not correct- “Performance measures” is typically a part of the exit criteria.

Question 32

The following diagram shows the logical dependencies between a set of nine requirements, where a dependency is shown by an arrow. For example, “R3 → R6” means that R6 depends on R3.



Which one of the following options structures the test execution schedule according to the requirement dependencies?

- a) R1 → R3 → R5 → R2 → R4 → R7 → R6 → R8 → R9.
- b) R1 → R3 → R5 → R2 → R4 → R3 → R8 → R6 → R4 → R7 → R6 → R9.
- c) R5 → R2 → R5 → R1 → R3 → R4 → R3 → R7 → R6 → R8 → R9.
- d) **R5 → R1 → R2 → R4 → R3 → R7 → R8 → R6 → R9.**

FL-5.2.4 (K3) Apply knowledge of prioritization, technical and logical dependencies to schedule test execution for a given set of test cases

Justification

- a) Not correct – everything is dependent on R5, so any test flow that does not start with R5 is FALSE.
- b) Not correct – everything is dependent on R5, so any test flow that does not start with R5 is FALSE.
- c) Not correct – R3 is dependent on R4, so R4 should be tested before R3.
- d) **Correct** – the tests are specified in a sequence that takes the dependencies into account.

Question 33

Which of the following BEST describes the task carried out by the testers in the project?

- a) Introduce suitable metrics for measuring test progress.
- b) Determine the test schedule.
- c) **Prepare and acquire test data.**
- d) Prepare and deliver test progress report to the stakeholders.

FL-5.1.2 (K1) Identify the tasks of a test manager and tester

Justification

- a) Not correct – This is the task of test manager. (Syllabus 5.1.2)
- b) Not correct – This is the task of test manager. (Syllabus 5.1.2)
- c) **Correct** – Preparing and acquiring test data to carry out testing activities is the task of the tester in the project. (Syllabus 5.1.2)
- d) Not correct – This is the task of test manager. (Syllabus 5.1.2)

Question 34

Which of the following metrics can be MOST useful to assess the current quality of the test object?

- a) Details of test cases run/not run.
- b) Percentage of work done in test environment preparation.
- c) **Details of defects found and fixed during test execution.**
- d) Cost of testing.

FL-5.3.1 (K1) Recall metrics used for testing

Justification

- a) Not correct – Details of executed test cases would be helpful to assess progress against the planned schedule, not the current quality. (Syllabus 5.3.1)
- b) Not correct – Percentage of work done in test environment preparation would be helpful to assess progress against the planned schedule, not the current quality. (Syllabus 5.3.1)
- c) **Correct** – Details of defects found and fixed during test execution will be helpful to assess the current quality of the test object. (Syllabus 5.3.1)
- d) Not correct – Cost of testing will be helpful to assess the effectiveness of the test approach, not the current quality. (Syllabus 5.3.1)

Question 35

Which of the following BEST describes the difference between the test progress report and test summary report?

- a) The test progress report is prepared for the internal stakeholders, whereas a test summary report is prepared for external stakeholders.
- b) The test progress report covers only the status of test activities, whereas a test summary report covers only the quality of the test object.
- c) **The test progress report is prepared at regular intervals after the entry criteria are reached, whereas a test summary report is prepared after the exit criteria is reached for testing.**
- d) The test progress report is prepared at the end of each test activity, whereas a test summary report is prepared during the test activity.

FL-5.3.2 (K2) Summarize the purposes, content, and audiences for test reports

Justification

- a) Not correct – Both reports are for the testing audience i.e. internal and external stakeholders. (Syllabus 5.3.2)
- b) Not correct – Both reports cover the status of test activities and quality of test object (Syllabus 5.3.2)
- c) **Correct** – The test progress report is prepared at regular intervals after the entry criteria is reached, while the test summary report is prepared once the exit criteria is reached for testing. (Syllabus 5.3.2).
- d) Not correct – The test progress report is prepared during the test activity, whereas a test summary report is prepared at the end of the test activity. (Syllabus 5.3.2)

Question 36

Which of the following is NOT a characteristic of the metrics-based test estimation technique?

- a) Project budget from similar projects in the past can be used for estimation.
- b) Defects data from similar projects in the past can be used for estimation.
- c) **Average of estimates collected from technical experts can be used for estimations.**
- d) Standard timeframe to set up a test environment from similar projects can be used for estimation.

FL-5.2.6 (K2) Explain the difference between two estimation techniques: the metrics-based technique and the expert-based technique

Justification

- a) Not correct – This is characteristic of metrics-based approach where the estimation of the test effort is based on the metrics of similar projects in the past or based on typical values.
- b) Not correct – This is characteristic of metrics-based approach where the estimation of the test effort is based on the metrics of similar projects in the past or based on typical values.
- c) **Correct** – This is characteristic of the expert-based approach where the estimation of the task is based on the estimates from the owners of the tasks or by experts. (Syllabus 5.2.6)
- d) Not correct – This is characteristic of metrics-based approach where the estimation of the test effort is based on metrics of similar projects in the past, or based on typical values.

Question 37

The project for the enhancement of a banking application is in the last phase of testing. During the regression testing, the tester has found that some of the customers are not able to generate the monthly account statement from the website. This functionality works fine from the mobile app for the same users.

Tester has written the defect report to highlight this issue with the following information:

Defect ID: 139

Title: Monthly account statement functionality not working

Summary: After the user clicks on the statement button on the account page, monthly account statement is not generated. The same users are able to generate the monthly account statement from the mobile app.

Severity of defect: High

Priority of defect: High

Expected result: All the users should be able to generate the monthly account statement.

Actual result: Some of the users are not able to generate the monthly account statement.

What is the MOST valuable information that is missed by the tester in the above defect report?

- a) Test case reference for this defect.
- b) Test environment and detected phase.
- c) Date and author of the defect report.
- d) **Data used for testing.**

FL-5.6.1 (K3) Write a defect report, covering defects found during testing.

Justification

- a) Not correct – This is not important as the details are already in the summary.
- b) Not correct – As described earlier in the question the project is in the testing phase so the detected environment and detected phase will be known to the developer. Moreover, this information is not much important in this situation.

- c) Not correct – This information is missing but not much important for this situation.
- d) **Correct** – As this behavior is observed only for some of the customers, details to the test data used is important for the developer to replicate the problem. Syllabus 5.6 - A description of the defect to enable reproduction and resolution, including logs, database dumps, screenshots or recordings.

Question 38

Which of the following statement describing the different type of test strategies is false?

- a) Model-based test strategy involves designing tests based on some model for a required aspect of the product.
- b) Consultative test strategy involves designing tests based on the advice, guidance, or instructions of stakeholders and business domain experts.
- c) **Analytical test strategy involves designing tests based on industry-specific standards, or by processes imposed by the organization.**
- d) Methodical test strategy involves designing tests based on some predefined set of test conditions or likely types of failures.

FL-5.2.2 (K2) Differentiate between various test strategies

Justification

- a) Not correct – This is true. This correctly describes the Model-based test strategy
- b) Not correct – This is true. This correctly describes the consultative test strategy.
- c) **Correct** – This is false. This describes the process-compliant or standard-compliant test strategy, not the analytical test strategy.
- d) Not correct – This is true. This correctly describes the Methodical test strategy.

Question 39

Which of the following cannot be considered as a valid objective of a pilot project while introducing a new tool into an organization?

- a) Assess whether the benefits will be achieved at a reasonable cost.
- b) Understand the strength and weakness of the tool.
- c) **To identify the tool owner who can implement the tool in the organization.**
- d) Evaluate how the tool fits in with the existing processes and practices.

FL-6.2.2 (K1) Recall the objectives for using pilot projects to introduce tools

Justification

- a) Not correct – This is one of the objectives of the pilot project. (Syllabus 6.2.2)
- b) Not correct – This is one of the objectives of the pilot project. (Syllabus 6.2.2)
- c) **Correct** – This is not a valid objective of the pilot project.
- d) Not correct – This is one of the objectives of the pilot project. (Syllabus 6.2.2)

Question 40

Which of the following tool is most useful for the early detection of integration problems and conflicting changes?

- a) Performance testing tool
- b) Continuous integration tool**
- c) Coverage tool
- d) Static analysis tool

FL-6.1.1 (K2) Classify test tools according to their purpose and the test activities they support

Justification

- a) Not correct – Performance testing tools are used to simulate a load on an application or a database, or a network or server to assess system performance.
- b) **Correct** – Continuous Integration (CI) tools allow integrating all changes made to the code regularly. This makes it easier to detect integration problems and conflicting changes.
- c) Not correct – Coverage tools are used to measure the percentage of specific types of code structures that have been exercised (e.g. statements, decisions, or function calls).
- d) Not correct – Static Analysis tools are used to calculate the complexity of code based on the number of decisions and some other factors.

4

Practice Exam 2

Question 1

Which of the following factors can influence the test process for an organization?

- I. Business domain
- II. Budgets and resources
- III. Software development lifecycle model
- IV. Product risks

- a) II, III, IV
- b) I, II, III
- c) I, III, IV
- d) All of the above

Select one option.

Question 2

Differentiate the following testing terms (1-4) by mapping them to the right examples (A-D).

1. Error
 2. Defect
 3. Failure
 4. Root cause
-
- A. The requirement is ambiguous due to the lack of knowledge on the part of the business analyst.
 - B. Incorrect interest is calculated for the customer accounts.
 - C. The program is using the wrong algorithm for interest calculation.
 - D. The developer misinterpreted the requirement for interest calculation.
-
- a) 1A, 2C, 3B, 4D.
 - b) 1D, 2B, 3A, 4C.
 - c) 1D, 2C, 3B, 4A.
 - d) 1A, 2C, 3D, 4B.

Select one option.

Question 3

Which TWO of the following statements describing the advantage of having traceability between the test basis and test work products are TRUE?

- a) Traceability helps to analyze the impact of new changes in software.
- b) Traceability helps to find the necessary test data to support test conditions and test cases.
- c) Traceability helps to analyze anomalies to establish their likely causes.
- d) Traceability helps to provide the necessary information to assess product quality.

Select two options.

Question 4

Which one of the following is the BEST description for test implementation?

- a) The activity that involves checking the status of test activities, to identify any variances from the planned or expected status, and reporting status to the stakeholders.
- b) The activity that derives and specifies test cases from test conditions.
- c) The activity that identifies test conditions by analyzing the test basis.
- d) The activity that prepares the testware needed for test execution based on test analysis and design.

Select one option.

Question 5

Which of the TWO below tasks are performed during the Test implementation activity of the ISTQB Test Process?

- a) Designing and prioritizing test cases.
- b) Capturing bi-directional traceability between the test basis, test conditions, test cases, and test procedures.
- c) Preparing test data and ensuring it is properly loaded in the test environment.
- d) Setting up the test environment.

Select two options.

Question 6

The following statements are related to the use of test techniques to reduce the chances of problematic delivery. Which of the following is FALSE?

- a) Testers can work closely with the developers to increase understanding of code and to reduce the risk of defects within the code and the tests.
- b) Testers can work closely with system designers to find out fundamental design defects at an early stage.
- c) Testers can work closely with the project manager to understand the project schedule which can help in designing the appropriate test to meet milestone dates.
- d) Testers can verify and validate the system prior to the release to increase the likelihood of software meeting stakeholders needs.

Select one option.

Question 7

Which of the following statement related to software testing principles is TRUE?

- I. If the regression tests are automated, they can still find new defects over a period of time.
 - II. If enough testing is done, still it cannot be proved that there are no more defects remaining.
 - III. If the testing is started early, it will increase the overall cost and schedule of projects.
 - IV. If testing is able to find a large number of defects, it can ensure the users' needs are fulfilled.
-
- a) I and II
 - b) Only II
 - c) II and III
 - d) Only I

Select one option.

Question 8

Which of the following is the BEST way to reduce the perception of testing as a destructive activity?

- a) Testers actively participating in the defect review meetings.
- b) Testers planning regular meetings with developers and business analysts.
- c) Testers providing defect information in a constructive way to the stakeholders.
- d) Testers confirming the defects with developers before creating defect reports.

Select one option.

Question 9

Which of the following statement comparing functional testing and non-functional testing is TRUE?

- 1. Functional testing evaluates the behavior of the system, whereas non-functional testing evaluates the functions that the system should perform.
 - 2. Functional testing uses black-box techniques whereas non-functional testing does not use black-box techniques.
 - 3. Functional testing is the testing of “what” the system should do, whereas non-functional testing is the testing of “how well” the system behaves.
 - 4. Functional testing can be performed at all test levels, whereas non-functional testing can be performed only at the system level.
-
- a) 2 and 3
 - b) Only 3
 - c) 1 and 4
 - d) All of them.

Select one option.

Question 10

Which TWO of the following statements comparing component integration testing and system integration testing are TRUE?

- a) Component integration testing verifies the functionality of software modules, whereas system integration testing verifies interfaces between components of the system.
- b) Component integration testing focuses on the interactions between integrated components, whereas system integration testing focuses on the interactions between systems.
- c) Component integration testing is often the responsibility of the developers, whereas system integration testing is often the responsibility of the testers.
- d) Component integration testing focuses only on functional characteristics, whereas system integration testing focuses on both functional and non-functional characteristics.

Select two options.

Question 11

Which of the following statement about the role of impact analysis in maintenance testing is FALSE?

- a) Impact analysis evaluates the side effects of the changes to the system.
- b) Impact analysis can be difficult if the requirements are out of date.
- c) Impact analysis is always done after the changes are made to the system.
- d) Impact analysis can help to identify the impact of a change on existing tests.

Select one option.

Question 12

Which of the following statements relates to different software development models?

1. Design, build and testing activities are often overlapping throughout the overall development cycle.
 2. The testing activities start once the development activities are completed.
 3. Each cycle delivers a working software which is a subset of the overall set of features.
 4. Specifying requirements, design, build and testing are done in a series with added pieces.
-
- a) 2 & 4 are related to Sequential and 1 & 3 are related to iterative and incremental development.
 - b) 1 & 2 are related to Sequential and 2 & 3 are related to iterative and incremental development.
 - c) 1 is related to Sequential and 2, 3 & 4 are related to iterative and incremental development.
 - d) 2 is related to Sequential and 1, 3 & 4 are related to iterative and incremental development.

Select one option.

Question 13

How can change-related testing be applied during system testing?

- a) Verify that 100% statements coverage is done for all paths.
- b) Verify that all workflow tests related to changes are re-executed.
- c) Verify that data can be successfully transferred between systems.
- d) Verify that data transferred between components is secure.

Select one option.

Question 14

Which of the following TWO statements describing differences between static and dynamic testing is TRUE?

- a) Static testing focuses on improving the internal quality of the work products, whereas dynamic testing typically focuses on externally visible behaviors.
- b) The objective of static testing is to find defects, whereas the objective of dynamic testing is to find failures.
- c) Dynamic testing can find the similar defects found during static testing but with much less effort.
- d) There are some static testing defects which cannot be found using dynamic testing.

Select two options.

Question 15

One of the banks is upgrading their legacy banking application. This will allow internal and external users to access the application through additional interfaces.

The expected users of the application are:

- Customers
- Tellers
- Customer care specialists

These users can use the following interfaces to login to the banking application.

Customer- Online browser or mobile app.

Teller – Stand-alone desktop application

Customer care specialist – Stand-alone desktop application and online browser

Different reviewers were requested to review the application from the perspective of the above user categories with the focus on ease of use.

Which of the following review comment is MOST LIKELY common for most of the reviewers?

- a) A lot of scrolling is required to see all the new offers page.
- b) The text size is too small for the user to read all the information on the small device.
- c) After login to the application, the customer search functionality is difficult to use.
- d) Changing all the customer details requires going through a lot of different pages.

Select one option.

Question 16

When are the following activities performed during a work product review process?

- Communicating identified potential defects
 - Potential defects analyzed and assigned to the owners
 - Evaluation of quality characteristics
-
- a) Individual review.
 - b) Initiate review.
 - c) Issue communication and analysis.
 - d) Fixing and reporting.

Select one option.

Question 17

Match the different Review types (1-4) with their BEST respective attributes (A-D) below?

1. Walkthrough.
 2. Technical Review.
 3. Informal Review.
 4. Inspection.
-
- A. Follows a defined process with formal documented outputs.
 - B. May take the form of scenarios or dry runs.
 - C. Reviewers should be technical experts.
 - D. May not involve a review meeting.
-
- a) 1D, 2B, 3A, 4C.
 - b) 1D, 2C, 3B, 4A.
 - c) 1B, 2C, 3A, 4D.
 - d) 1B, 2C, 3D, 4A.

Select one option.

Question 18

Which of the following is the responsibly of “Management” in a formal review?

- a) Takes overall responsibility for the review.
- b) Ensures effective running of the review meeting.
- c) Decides the participants for review.
- d) Decides on the execution of reviews.

Select one option.

Question 19

A computer manufacturing company is working on the marketing strategy for the launch of their new products. The customers are targeted based on the following characteristics:

- Use (personal or work related)
- Mobility
- Age group: kids (under 19): adults (between 19-65), seniors (over 65)

The following decision table has been designed to test the logic of marketing strategy:

Rules	R1	R2	R3	R4	R5	R6	R7	R8
Conditions								
Personal	Y	N	Y	N	N/A	N/A	N/A	N/A
Mobility	N	N	Y	Y	N	Y	N	Y
Kids	N	N	N	N	Y	Y	N	N
Adults	Y	Y	Y	Y	N	N	N	N
Seniors	N	N	N	N	N	N	Y	Y
Actions								
Desktop	Y	Y	N	N	N	N	N	N
Laptop	Y	N	Y	Y	N	N	Y	N
Normal Tablet	N	N	Y	N	Y	Y	Y	Y
Mini Tablet	N	N	N	N	Y	Y	N	N

According to the company's marketing strategy which products are suitable for the following customer?

- Sandra is 19-year-old, looking for a product for work use without mobility.
 - Andrew is 70-year-old, looking for a product for personal use with mobility.
- Sandra- Normal Tablet or Mini Tablet; Andrew- Normal Tablet.
 - Sandra- Desktop; Andrew- Laptop and Normal Tablet.
 - Sandra- Desktop or laptop; Andrew- Normal Tablet.
 - Sandra- Desktop; Andrew- Normal Tablet.

Select one option

Question 20

Which of the following BEST describes the difference between statement testing and decision testing?

- a) Statement testing covers all the positive paths in the code, whereas decision testing covers all the positive and negative paths in the code.
- b) Statement testing is best suited for an application having simple logic, whereas decision testing is best suited for an application with complex business rules.
- c) Statement testing ensures that all potential executable statements in the code have been tested, whereas decision testing ensures that all decision logic has been tested.
- d) Statement testing is a white-box testing technique, whereas decision testing is a black-box testing technique.

Select one option.

Question 21

A motor toll system calculates the one-way toll amount based on the type of the vehicle.

- For car, the toll is \$10
- For van, the toll is \$10
- For taxi, the toll is \$15
- For truck the toll is \$20

Motorbikes are exempted from any toll. Calculate the minimum number of test cases by applying the equivalence partitioning test technique to test this functionality?

- a) 6 tests.
- b) 3 tests.
- c) 4 tests.
- d) 5 tests.

Select one option.

Question 22

Consider the following statement:

“The code has two IF statements which are independent of each other, with no other loops or case statements. Only two tests are able to provide 100% decision coverage with each test providing 50% decision coverage.

Which of the following is correct with respect to the above scenario?

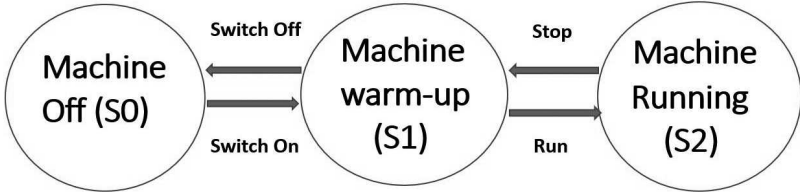
- a) This is possible; however, if any of the IF statements have an executable statement for false outcome then the number of tests will change.
- b) This is not possible; there are four decision outcomes so a single test can provide a maximum of 25% decision coverage.
- c) This is possible; for two independent IF statements two tests can provide 100% decision coverage.
- d) This is not possible; for two independent IF statements more than two tests are required for 50% decision coverage.

Select one option.

Question 23

Consider a system for the treadmill machine used in the gym. The treadmill machine has to be in the warm-up state before it can be used for running. While the machine is in running state it cannot be switched-off.

The state transition diagram and table of test cases are shown below:



Test case	1	2	3	4	5	6
Start state	S0	S1	S1	S2	S2	S2
Input	Switch off	Switch on	Stop	Switch on	Switch off	Run
Expected final state	S0	S1	S1	S2	S2	S2

Which of the following statements are true about the given state transition diagram and table of test cases?

- The given test cases represent all the critical transitions in the state transition diagram.
- The given test cases can be used to cover some of the valid and invalid transitions in the state transition diagram.
- The given test cases represent some of the invalid transitions in the state transition diagram. More tests are required to cover all the invalid transactions.
- The given test cases represent all possible invalid transitions in the state transition diagram.

Select one option.

Question 24

A leading mobile company is enhancing some of the features of an existing phone application. Existing business process diagrams defining the user interaction with the phone are updated based on the changes.

Which of the following test technique will be BEST suited for deriving test cases?

- a) State transition testing.
- b) Decision table testing.
- c) Checklist-based testing.
- d) Use-case testing.

Select one option.

Question 25

The bank ATM machine charges withdrawal fees for the other bank's card. The fee is calculated based on the amount withdrawn:

- For amount \$500 or less the fee is \$1.
- For amount over \$500 and up to \$1000, the fee is \$2.
- For any amount more than \$1000 the fee is \$3.

How many tests will be required to test all set of values (\$) for three-point boundary value analysis? The withdrawal amounts can only be in multiples of \$1.

- a) 10
- b) 9
- c) 8
- d) 7

Select one option.

Question 26

Which of the following statement regarding exploratory testing is TRUE?

- a) Exploratory testing requires some of the tests to be pre-defined and rest are designed dynamically during execution.
- b) Exploratory testing is not effective if the system is already tested thoroughly using formal testing techniques.
- c) Exploratory testing can provide higher coverage than other formal testing techniques if the testers are having strong experience in similar applications.
- d) Exploratory testing is more effective when it is performed within a defined time-box using test charters.

Select one option.

Question 27

What is use case testing?

- a) A test technique in which tests are executed based on the tester's experience, knowledge and intuition.
- b) A test technique in which state table is used to derive test cases to validate valid and invalid transitions.
- c) A test technique in which test cases are derived based on the analysis of the internal structure of the system.
- d) A test technique in which test cases are designed to execute scenarios of use cases.

Select one option.

Question 28

While using the experience-based test technique for testing, which of the following TWO information would be MOST helpful?

- a) Knowledge of the system environment.
- b) knowledge of the internal structure of the system.
- c) Knowledge of the expected use of the software.
- d) Knowledge about the code of the test object.

Select two options.

Question 29

Consider a system built to automatically calculate the government stamp duty based on the purchase price of the house. There is no stamp duty for houses where the purchase price is less than \$200,000. The maximum stamp duty is 3.5% which is applied for houses where the purchase price is more than or equal to \$800,000.

Rest of the rules for calculating stamp duty are:

- 1.0% for houses with purchase price from \$200,000 and less than \$400,000
- 1.5% for houses with purchase price from \$400,000 and less than \$500,000
- 2.0% for houses with purchase price from \$500,000 and less than \$600,000
- 2.5% for houses with purchase price from \$600,000 and less than \$700,000
- 3.0% for houses with purchase price from \$700,000 and less than \$800,000

What is the minimum number of test cases required to cover all valid equivalence partitions for calculating the stamp duty?

- a) 5
- b) 8
- c) 6
- d) 7

Select one option.

Question 30

For the following tests, organize the different degrees of test independence from the low level to high level.

- I. Tests designed by the testers from the same team.
 - II. Tests designed by the test specialist from an independent test team.
 - III. Tests designed by the developer who wrote the code.
 - IV. Tests designed by another developer from the development team.
 - V. Tests designed by the tester from the outsourced company.
- a) (III), (IV), (II), (I), (V)
 - b) (III), (IV), (I), (V), (II)
 - c) (III), (IV), (I), (II), (V)
 - d) (V), (II), (I), (IV), (III)

Select one option.

Question 31

Which of the following statements regarding the entry and exit criteria for testing is FALSE?

- a) Exit criteria define what conditions must be achieved in order to declare a test level complete.
- b) Entry criteria define the preconditions for undertaking a given test activity.
- c) Typical exit criteria include: Reliability measures, test cost, testable code ready, time to market, and product completeness.
- d) Typical exit criteria include: Performance efficiency, residual risks and code coverage.

Select one option.

Question 32

Which of the following statements correctly describe the tasks of the test manager and the tester?

- a) The test manager develops or reviews the test strategy to be followed, whereas the tester writes and updates the test plan.
- b) The test manager initiates the analysis, design, implementation, and execution of tests, whereas the tester prepares and delivers the test progress reports and test summary reports.
- c) The tester designs and implements test cases, whereas the test manager review tests developed by testers.
- d) The test manager decides about the implementation of the test environment, whereas the tester designs, set up and verifies the test environment.

Select one option.

Question 33

Which of the following correctly describes the difference between metrics-based and expert-based techniques for test estimation?

- a) Metrics-based approach is applicable only for sequential development model, whereas expert-based approach is applicable for any development model.
- b) Metrics-based approach accuracy depends on the quantity of metrics, whereas expert-based approach accuracy depends on the years of experience of experts in their respective fields.
- c) Metrics-based approach can be used if there is sufficient data from previous similar projects, whereas expert-based approach requires task experts for estimation.
- d) Metrics-based approach can be used for all test level estimations, whereas expert-based approach is applicable for system level test estimations.

Select one option.

Question 34

Given the following priorities and dependencies for the test cases created for a banking system:

Test ID	Test case Name	Dependency	Priority
TC1.	Login	None	Low
TC2.	Account Summary	TC1	High
TC3.	Transfer Payment	TC2	Medium
TC4.	Add Payee	TC3	High
TC5.	Check Statement	TC2	Medium
TC6.	Update Details	TC2	High

Which of the following test execution schedules BEST considers the priorities and dependencies?

- a) TC2 → TC1 → TC3 → TC4 → TC5 → TC6
- b) TC1 → TC2 → TC4 → TC6 → TC3 → TC5.
- c) TC1 → TC2 → TC6 → TC3 → TC5 → TC4.
- d) TC1 → TC2 → TC6 → TC3 → TC4 → TC5.

Select one option.

Question 35

Which of the following metrics would be MOST useful to measure progress during the test execution?

- a) Percentage of test conditions passed/failed.
- b) Percentage of planned test cases implemented.
- c) Details of defects closed during test execution.
- d) Resource allocation and usage during test execution.

Select one option.

Question 36

Which of the following should be included as a part of the test progress report?

- I. The status of the test activities and progress against the test plan.
- II. Factors impeding the test progress.
- III. Lessons learned from the completed test activities.
- IV. Testing activities planned for the next reporting period.
- V. The current quality of the test object.

- a) I, II
- b) I, II, IV
- c) I, II, IV, V
- d) All of the above

Select one option.

Question 37

Which of the following statement correctly describes the difference between analytical and reactive test strategy for testing?

- a) Analytical test strategy is based on the analysis of risk, whereas reactive test strategy is based on the instructions of the business domain experts.
- b) Analytical test strategy implements tests based on external rules and standards, whereas in reactive test strategy tests are implemented based on the prior test results.
- c) In analytical test strategy, tests and their order are pre-planned based on risk analysis, whereas in reactive test strategy tests are not pre-planned but are created based on the prior tests results.
- d) Analytical test strategy uses extensive automated regression tests, whereas reactive test strategy uses extensive exploratory tests.

Select one option.

Question 38

A new tester in the project is responsible for testing the front end of the system which will be used by the internal staff. There are two test users which can be used for testing the internal staff functionality.

While testing the user detail page, the tester has observed that static text for all the input fields is starting with lowercase letters. For both the test users the observation is similar.

In the specification document, it was clearly mentioned that all the static fields should start with an uppercase letter.

Tester has written the defect report to highlight this issue with the following information:

Defect ID: 1009

Title: Static text starts with lowercase in user detail page

Short summary: After the internal user login to the front-end application, the user detail page shows all the static fields in lowercase letters.

Expected result: All the static fields should start with an uppercase letter.

What is the MOST important piece of information that is missed by the tester in the above defect report?

- a) Test data used for testing.
- b) The actual test results.
- c) Severity and priority of the defect.
- d) Date and author of the defect

Select one option.

Question 39

Which of the following should be BEST considered while successfully implementing a new tool within an organization?

- a) The organization should not plan to start any test activity unless the complete rollout for new tool is completed.
- b) The organization should plan for complete rollout of the tool in one go rather than incrementally.
- c) The organization should plan to make changes to its development lifecycle to fit with the use of the tool.
- d) The organization should plan that processes should be adapted and improved to fit with the use of the tool.

Select one option.

Question 40

Which of the following is a characteristic of a test harness tool?

- a) It helps to enable tests for automatic/semi-automatic execution using stored inputs.
- b) It helps to store information about versions and builds of software and testware.
- c) It helps to provide a predictable and controllable environment in which the faults can be localized.
- d) It helps to find inconsistencies in the database model.

Select one option.

Answers

1	D	K2
2	C	K2
3	A, D	K2
4	D	K1
5	C, D	K2
6	C	K2
7	B	K2
8	C	K1
9	B	K2
10	B, C	K2
11	C	K2
12	D	K2
13	B	K1
14	A, D	K2
15	D	K3
16	C	K2
17	D	K2
18	D	K1
19	D	K3
20	C	K2

21	D	K3
22	C	K2
23	C	K3
24	D	K2
25	C	K3
26	D	K2
27	D	K1
28	A, C	K2
29	D	K3
30	C	K2
31	C	K2
32	D	K1
33	C	K2
34	D	K3
35	A	K1
36	C	K2
37	C	K2
38	C	K3
39	D	K1
40	C	K2

5

Answers – Practice Exam 2

Question 1

Which of the following factors can influence the test process for an organization?

- I. Business domain
- II. Budgets and resources
- III. Software development lifecycle model
- IV. Product risks

- a) II, III, IV
- b) I, II, III
- c) I, III, IV
- d) **All of the above**

FL-1.4.1 (K2) Explain the impact of context on the test process

Justification

- I. The business domain can influence the test process for an organization. Syllabus 1.4.1
- II. Operational constraints such as budgets and resources can influence the test process for an organization. Syllabus 1.4.1
- III. Software development lifecycle model and project methodologies can influence the test process for an organization. Syllabus 1.4.1
- IV. Product risks can influence the test process for an organization. Syllabus 1.4.1

Thus:

- a) Not correct
- b) Not correct
- c) Not correct
- d) **Correct**

Question 2

Differentiate the following testing terms (1-4) by mapping them to the right examples (A-D).

- | | |
|-----------|---------------|
| 1. Error | 3. Failure |
| 2. Defect | 4. Root cause |
-
- A. The requirement is ambiguous due to the lack of knowledge on the part of the business analyst.
- B. Incorrect interest is calculated for the customer accounts.
- C. The program is using the wrong algorithm for interest calculation.
- D. The developer misinterpreted the requirement for interest calculation.
-
- | | |
|--------------------|---------------------------|
| a) 1A, 2C, 3B, 4D. | c) 1D, 2C, 3B, 4A. |
| b) 1D, 2B, 3A, 4C. | d) 1A, 2C, 3D, 4B. |

FL-1.2.3 (K2) Distinguish between error, defect, and failure

Justification

An error is a mistake by a person which can lead to the introduction of the defect. In this case, the developer has made the mistake which resulted in the wrong code. **(1D)**

The defect is an imperfection or deficiency in a work product or code where it does not meet the requirements or specifications. In this case, the program using the wrong algorithm for interest calculation is a defect. **(2C)**

Failure is an event in which a component or system does not perform a required function within specified limits. In this case, it is the incorrect interest calculated for the customers. **(3B)**

The root causes of defects are the earliest actions or conditions that contributed to creating the defects. In this case, it was the lack of knowledge on the part of the business analyst for the ambiguous requirement. **(4A)**

Thus:

- a) Not correct
- b) Not correct
- c) **Correct**
- d) Not correct

Question 3

Which TWO of the following statements describing the advantage of having traceability between the test basis and test work products are TRUE?

- a) **Traceability helps to analyze the impact of new changes in software.**
- b) Traceability helps to find the necessary test data to support test conditions and test cases.
- c) Traceability helps to analyze anomalies to establish their likely causes.
- d) **Traceability helps to provide the necessary information to assess product quality.**

FL-1.4.4 (K2) Explain the value of maintaining traceability between the test basis and test work Products

Justification

- a) **Correct** – This is true. Traceability supports in analyzing the impact of changes. Syllabus 1.4.4.
- b) Not correct – This is false. This test activity is not related to Traceability.
- c) Not correct – This is false. This test activity is not related to Traceability.
- d) **Correct** – This is true. Traceability provides information to assess product quality, process capability, and project progress against business goals. Syllabus 1.4.4.

Question 4

Which one of the following is the BEST description for test implementation?

- a) The activity that involves checking the status of test activities, to identify any variances from the planned or expected status, and reporting status to the stakeholders.
- b) The activity that derives and specifies test cases from test conditions.
- c) The activity that identifies test conditions by analyzing the test basis.
- d) The activity that prepares the testware needed for test execution based on test analysis and design.**

FL-1.x (K1) Keywords Chapter 1**Justification**

- a) Not correct – This is the definition of test monitoring according to Glossary.
- b) Not correct – This is the definition of test design according to Glossary.
- c) Not correct – This is the definition of test analysis according to Glossary
- d) **Correct** – This is the definition of test implementation from Glossary.

Question 5

Which of the TWO below tasks are performed during the Test implementation activity of the ISTQB Test Process?

- a) Designing and prioritizing test cases.
- b) Capturing bi-directional traceability between the test basis, test conditions, test cases, and test procedures.
- c) **Preparing test data and ensuring it is properly loaded in the test environment.**
- d) **Setting up the test environment.**

FL-1.4.2 (K2) Describe the test activities and respective tasks within the test process

Justification

- a) Not correct – this activity is performed during the Test design activity.
- b) Not correct – this activity is performed during the Test design activity.
- c) **Correct** – this activity is performed during the Test implementation activity. Syllabus 1.4.2.
- d) **Correct** – this activity is performed during the Test implementation activity. Syllabus 1.4.2.

Question 6

The following statements are related to the use of test techniques to reduce the chances of problematic delivery. Which of the following is FALSE?

- a) Testers can work closely with the developers to increase understanding of code and to reduce the risk of defects within the code and the tests.
- b) Testers can work closely with system designers to find out fundamental design defects at an early stage.
- c) **Testers can work closely with the project manager to understand the project schedule which can help in designing the appropriate test to meet milestone dates.**
- d) Testers can verify and validate the system prior to the release to increase the likelihood of software meeting stakeholders needs.

FL-1.2.1 (K2) Give examples of why testing is necessary**Justification**

- a) Not correct – This is true. Testers can work closely with the developers to get a good understanding of the code and how to test it. This increased understanding can reduce the risk of defects within the code and the tests. (Syllabus 1.2.1)
- b) Not correct – This is true. Testers can work closely with the system designers to get a good understanding of the design and how to test it, this increased understanding can reduce the risk of design defects and enable tests to be identified at an early stage. (Syllabus 1.2.1)
- c) **Correct**– This is false. Tests are not designed to meet the milestone dates. Moreover, this approach is not effective in reducing the frequency of problematic delivery.
- d) Not correct – This is true. Testers can verify and validate the software prior to release. This can detect failures that might otherwise have been missed. This also increases the likelihood that the software meets stakeholder needs and satisfies requirements. (Syllabus 1.2.1)

Question 7

Which of the following statement related to software testing principles is TRUE?

- I. If the regression tests are automated, they can still find new defects over a period of time.
 - II. If enough testing is done, still it cannot be proved that there are no more defects remaining.
 - III. If the testing is started early, it will increase the overall cost and schedule of projects.
 - IV. If testing is able to find a large number of defects, it can ensure the users' needs are fulfilled.
- a) I and II
 - b) Only II**
 - c) II and III
 - d) Only I

FL-1.3.1 (K2) Explain the seven testing principles

Justification

- This is contradicting Principle#5. Same tests (automated or manual) repeated over and over again can't find defects. (Syllabus 1.3)
- This is true. Principle#1 -Testing can show that defects are present, but cannot prove that there are no more defects. (Syllabus 1.3)
- This is contradicting Principle#3. Testing early in the software development lifecycle helps reduce or eliminate costly changes. (Syllabus 1.3)
- This is contradicting Principle#7. Finding and fixing a large number of defects can't ensure the success of a system as it may not be able to fulfill the users' needs and expectations. (Syllabus 1.3)

Thus:

- a) Not correct
- b) Correct**
- c) Not correct
- d) Not correct

Question 8

Which of the following is the BEST way to reduce the perception of testing as a destructive activity?

- a) Testers actively participating in the defect review meetings.
- b) Testers planning regular meetings with developers and business analysts.
- c) **Testers providing defect information in a constructive way to the stakeholders.**
- d) Testers confirming the defects with developers before creating defect reports.

FL-1.5.1 (K1) Identify the psychological factors that influence the success of testing

Justification

- a) Not correct – This is required for testers but no way reduces the perception of testing as a destructive activity.
- b) Not correct – Regular meetings are required between testers and other team members but they do not reduce the perception of testing as a destructive activity.
- c) **Correct** – Information about defects and failures should be communicated in a constructive way to reduce the perception of testing as a destructive activity. (Syllabus 1.5.1)
- d) Not correct – Confirmation with developers is not required before creating the defect reports. This also, does not reduce the perception of testing as a destructive activity.

Question 9

Which of the following statement comparing functional testing and non-functional testing is TRUE?

1. Functional testing evaluates the behavior of the system, whereas non-functional testing evaluates the functions that the system should perform.
 2. Functional testing uses black-box techniques whereas non-functional testing does not use black-box techniques.
 3. Functional testing is the testing of “what” the system should do, whereas non-functional testing is the testing of “how well” the system behaves.
 4. Functional testing can be performed at all test levels, whereas non-functional testing can be performed only at the system level.
- a) 2 and 3
b) Only 3
 c) 1 and 4
 d) All of them.

Select one option.

FL-2.3.1 (K2) Compare functional, non-functional, and white-box testing

Justification

1. This is false. The first part of the statement is true and the second part is false. It is the functional tests that evaluate the functions that the system should perform (Syllabus 2.3.1, 2.3.2)
2. This is false. Black-box techniques can be applied to both functional and non-functional testing (Syllabus 2.3.1, 2.3.2)
3. This is true. Functional testing focuses on “what” the system should do and non-functional testing focuses on “how well” the system behaves. (Syllabus 2.3.1, 2.3.2)
4. This is false. Functional and non-functional testing can be performed at all test levels (Syllabus 2.3.1, 2.3.2)

Thus:

- a) Not correct
b) Correct
 c) Not correct
 d) Not correct

Question 10

Which TWO of the following statements comparing component integration testing and system integration testing are TRUE?

- a) Component integration testing verifies the functionality of software modules, whereas system integration testing verifies interfaces between components of the system.
- b) Component integration testing focuses on the interactions between integrated components, whereas system integration testing focuses on the interactions between systems.**
- c) Component integration testing is often the responsibility of the developers, whereas system integration testing is often the responsibility of the testers.**
- d) Component integration testing focuses only on functional characteristics, whereas system integration testing focuses on both functional and non-functional characteristics.

FL-2.2.1 (K2) Compare the different test levels from the perspective of objectives, test basis, test objects, typical defects and failures, and approaches and responsibilities

Justification

- a) Not correct – First part describes component testing not component integration testing whereas the second part describes component integration testing not system integration testing.
- b) Correct** – Component integration testing focuses on the interactions between different components, whereas System integration testing focuses on the interactions between different systems. **(Syllabus 2.2.1, 2.2.2)**
- c) Correct** – Component integration is typically done by developers, whereas system integration testing is typically performed by testers. **(Syllabus 2.2.1, 2.2.2)**
- d) Not correct – Both component integration and system integration focus on functional and non-functional characteristics.

Question 11

Which of the following statement about the role of impact analysis in maintenance testing is FALSE?

- a) Impact analysis evaluates the side effects of the changes to the system.
- b) Impact analysis can be difficult if the requirements are out of date.
- c) **Impact analysis is always done after the changes are made to the system.**
- d) Impact analysis can help to identify the impact of a change on existing tests.

FL-2.4.2 (K2) Describe the role of impact analysis in maintenance testing

Justification

- a) Not correct – this is true. Impact analysis identifies the intended consequences and possible side effects of the changes to the system. (Syllabus 2.4.2)
- b) Not correct – this is true. Impact analysis can be difficult if the specifications are out of date or missing. (Syllabus 2.4.2)
- c) **Correct** – this is false. Impact analysis can be done before a change is made to know the potential consequences in other areas of the system. This will help in deciding whether the changes should be made.
- d) Not correct – this is true. Impact analysis identifies the areas in the system that will be affected by the change and also help to identify the impact of a change on existing tests. (Syllabus 2.4.2)

Question 12

Which of the following statements relates to different software development models?

1. Design, build and testing activities are often overlapping throughout the overall development cycle.
 2. The testing activities start once the development activities are completed.
 3. Each cycle delivers a working software which is a subset of the overall set of features.
 4. Specifying requirements, design, build and testing are done in a series with added pieces.
- a) 2 & 4 are related to Sequential and 1 & 3 are related to iterative and incremental development.
 - b) 1 & 2 are related to Sequential and 2 & 3 are related to iterative and incremental development.
 - c) 1 is related to Sequential and 2, 3 & 4 are related to iterative and incremental development.
 - d) **2 is related to Sequential and 1, 3 & 4 are related to iterative and incremental development.**

FL-2.1.1 (K2) Explain the relationship between software development activities and test activities in the software life cycle

Justification

1. This is related to the **Iterative model**. Syllabus 2.1.1
2. This is related to the **waterfall model** (sequential development). Syllabus 2.1.1
3. This is related to the **Iterative model**. Syllabus 2.1.1
4. This is related to the **Incremental model**. Syllabus 2.1.1

Thus:

- a) Not correct
- b) Not correct
- c) Not correct
- d) **Correct**

Question 13

How can change-related testing be applied during system testing?

- a) Verify that 100% statements coverage is done for all paths.
- b) Verify that all workflow tests related to changes are re-executed.**
- c) Verify that data can be successfully transferred between systems.
- d) Verify that data transferred between components is secure.

FL-2.3.2 (K1) Recognize that functional, non-functional and white-box tests occur at any test level

Justification

- a) Not correct – This is a white-box test for component acceptance testing.
- b) **Correct** – For system testing, all tests for a given workflow should be re-executed for any workflow related changes. (Syllabus 2.3.5)
- c) Not correct – This is a white-box test for system integration testing.
- d) Not correct – This is a non-functional test for component integration testing.

Question 14

Which of the following TWO statements describing differences between static and dynamic testing is TRUE?

- a) **Static testing focuses on improving the internal quality of the work products, whereas dynamic testing typically focuses on externally visible behaviors.**
- b) The objective of static testing is to find defects, whereas the objective of dynamic testing is to find failures.
- c) Dynamic testing can find the similar defects found during static testing but with much less effort.
- d) **There are some static testing defects which cannot be found using dynamic testing.**

FL-3.1.3 (K2) Explain the difference between static and dynamic techniques, considering objectives, types of defects to be identified, and the role of these techniques within the software lifecycle

Justification

- a) **Correct** – This is true. Static testing can be used to improve the consistency and internal quality of work products, while dynamic testing typically focuses on externally visible behaviors. (Syllabus 3.1.3)
- b) Not correct – This is false. Both static and dynamic testing have the same objective to find defects as soon as possible. (Syllabus 3.1.3)
- c) Not correct – This is false. Defects found using static testing are easier to find and fix. (Syllabus 3.1.3)
- d) **Correct** – This is true. Most of the maintainability defects can be found **only** by static testing. (Syllabus 3.1.3)

Question 15

One of the banks is upgrading their legacy banking application. This will allow internal and external users to access the application through additional interfaces.

The expected users of the application are:

- Customers
- Tellers
- Customer care specialists

These users can use the following interfaces to login to the banking application.

Customer- Online browser or mobile app.

Teller – Stand-alone desktop application

Customer care specialist – Stand-alone desktop application and online browser

Different reviewers were requested to review the application from the perspective of the above user categories with the focus on ease of use.

Which of the following review comment is MOST LIKELY common for most of the reviewers?

- a) A lot of scrolling is required to see all the new offers page.
- b) The text size is too small for the user to read all the information on the small device.
- c) After login to the application, the customer search functionality is difficult to use.
- d) **Changing all the customer details requires going through a lot of different pages.**

FL-3.2.4 (K3) Apply a review technique to a work product to find defects

Justification

- a) Not correct – this will likely impact only the customers as they will get new offers and not the other users. So, the comment would come from the review that took the perspective of the customer, but not from other reviewers.

- b) Not correct – this comment would come from the review that took the perspective of the customer using mobile app, but not from other reviewers.
- c) Not correct – this comment would come from the review that took the perspective of the teller and customer care specialist, but not from other reviewers.
- d) **Correct** – Each type of user would check the ease of changing customer information.

Question 16

When are the following activities performed during a work product review process?

- Communicating identified potential defects
 - Potential defects analyzed and assigned to the owners
 - Evaluation of quality characteristics
- a) Individual review.
 - b) Initiate review.
 - c) **Issue communication and analysis.**
 - d) Fixing and reporting.

FL-3.2.1 (K2) Summarize the activities of the work product review process

Justification

- a) Not correct – See answer c).
- b) Not correct – See answer c).
- c) **Correct** – All of these activities are performed during **Issue communication and analysis** activity. (Syllabus 3.2.1)
- d) Not correct – See answer c).

Question 17

Match the different Review types (1-4) with their BEST respective attributes (A-D) below?

1. Walkthrough.
 2. Technical Review.
 3. Informal Review.
 4. Inspection.
-
- A. Follows a defined process with formal documented outputs.
 - B. May take the form of scenarios or dry runs.
 - C. Reviewers should be technical experts.
 - D. May not involve a review meeting.
-
- a) 1D, 2B, 3A, 4C.
 - b) 1D, 2C, 3B, 4A.
 - c) 1B, 2C, 3A, 4D.
 - d) **1B, 2C, 3D, 4A.**

FL-3.2.3 (K2) Explain the differences between different review types: informal review, walkthrough, technical review and inspection

Justification

1. **Walkthrough** may take the form of scenarios, dry runs, or simulations. (Syllabus 3.2.3) **(1B)**
2. **Technical Review** requires reviewers to be technical expert. (Syllabus 3.2.3) **(2C)**
3. **Informal Review** may not involve a review meeting (Syllabus 3.2.3) **(3D)**
4. **Inspection** follows a defined process with formal documented outputs (Syllabus 3.2.3) **(4A)**

Thus:

- a) Not correct
- b) Not correct
- c) Not correct
- d) **Correct**

Question 18

Which of the following is the responsible of “Management” in a formal review?

- a) Takes overall responsibility for the review.
- b) Ensures effective running of the review meeting.
- c) Decides the participants for review.
- d) **Decides on the execution of reviews.**

FL-3.2.2 (K1) Recognize the different roles and responsibilities in a formal review

Justification

- a) Not correct- This is the responsibility of Review leader. (Syllabus 3.2.2)
- b) Not correct- This is the responsibility of Facilitator. (Syllabus 3.2.2)
- c) Not correct- This is the responsibility of Review leader. (Syllabus 3.2.2)
- d) **Correct** – Management decides on the execution of reviews. (Syllabus 3.2.2)

Question 19

A computer manufacturing company is working on the marketing strategy for the launch of their new products. The customers are targeted based on the following characteristics:

- Use (personal or work related)
- Mobility
- Age group: kids (under 19): adults (between 19-65), seniors (over 65)

The following decision table has been designed to test the logic of marketing strategy:

Rules	R1	R2	R3	R4	R5	R6	R7	R8
Conditions								
Personal	Y	N	Y	N	N/A	N/A	N/A	N/A
Mobility	N	N	Y	Y	N	Y	N	Y
Kids	N	N	N	N	Y	Y	N	N
Adults	Y	Y	Y	Y	N	N	N	N
Seniors	N	N	N	N	N	N	Y	Y
Actions								
Desktop	Y	Y	N	N	N	N	N	N
Laptop	Y	N	Y	Y	N	N	Y	N
Normal Tablet	N	N	Y	N	Y	Y	Y	Y
Mini Tablet	N	N	N	N	Y	Y	N	N

According to the company's marketing strategy which products are suitable for the following customer?

- Sandra is 19-year-old, looking for a product for work use without mobility.
 - Andrew is 70-year-old, looking for a product for personal use with mobility.
- Sandra- Normal Tablet or Mini Tablet; Andrew- Normal Tablet.
 - Sandra- Desktop; Andrew- Laptop and Normal Tablet.
 - Sandra- Desktop or laptop; Andrew- Normal Tablet.
 - Sandra- Desktop; Andrew- Normal Tablet.**

FL-4.2.3 (K3) Apply decision table testing to derive test cases from given requirements

Justification

Sandra falls under **Rule R2** hence **desktop** is best suited for her.

Andrew falls under **Rule R8** hence **Normal Tablet** is best suited for him.

Hence

- a) Not correct- see explanation above.
- b) Not correct- see explanation above.
- c) Not correct- See explanation above.
- d) **Correct.**

Question 20

Which of the following BEST describes the difference between statement testing and decision testing?

- a) Statement testing covers all the positive paths in the code, whereas decision testing covers all the positive and negative paths in the code.
- b) Statement testing is best suited for an application having simple logic, whereas decision testing is best suited for an application with complex business rules.
- c) **Statement testing ensures that all potential executable statements in the code have been tested, whereas decision testing ensures that all decision logic has been tested.**
- d) Statement testing is a white-box testing technique, whereas decision testing is a black-box testing technique.

FL-4.3.1 (K2) Explain statement coverage

Justification

- a) Not correct – Statement testing covers all executable statements and decision testing covers all the decision outcomes and not positive or negative paths.
- b) Not correct –Statement and decision testing are both applicable for all the applications (simple logic or complex rules application).
- c) **Correct** – Statement testing ensures that all potential executable statements in the code have been tested whereas decision testing ensures that all decision logic has been tested. (Syllabus 4.3.1,4.3.2)
- d) Not correct – Both are white-box testing techniques.

Question 21

A motor toll system calculates the one-way toll amount based on the type of the vehicle.

- For car, the toll is \$10
- For van, the toll is \$10
- For taxi, the toll is \$15
- For truck the toll is \$20

Motorbikes are exempted from any toll. Calculate the minimum number of test cases by applying the equivalence partitioning test technique to test this functionality?

- a) 6 tests.
- b) 3 tests.
- c) 4 tests.
- d) 5 tests.

FL-4.2.1 (K3) Apply equivalence partitioning to derive test cases from given requirements

Justification

When the values are discrete, each enumeration value is an equivalence class itself. Therefore, the input partitions are:

1. Car
2. Van
3. Taxi
4. Truck
5. Motorbike

We need minimum one test to test each input partition by applying Equivalence Partitioning test technique. Each test case for these input partitions will also cover all the output partitions (\$0, \$10, \$15, \$20)

Therefore, five tests are sufficient to test all the valid partitions.

- a) Not correct – see explanation above.
- b) Not correct – see explanation above.
- c) Not correct – see explanation above.
- d) **Correct** – see explanation above.

Question 22

Consider the following statement:

“The code has two IF statements which are independent of each other, with no other loops or case statements. Only two tests are able to provide 100% decision coverage with each test providing 50% decision coverage.

Which of the following is correct with respect to the above scenario?

- a) This is possible; however, if any of the IF statements have an executable statement for false outcome then the number of tests will change.
- b) This is not possible; there are four decision outcomes so a single test can provide a maximum of 25% decision coverage.
- c) **This is possible; for two independent IF statements two tests can provide 100% decision coverage.**
- d) This is not possible; for two independent IF statements more than two tests are required for 50% decision coverage.

FL-4.3.2 (K2) Explain decision coverage

Justification

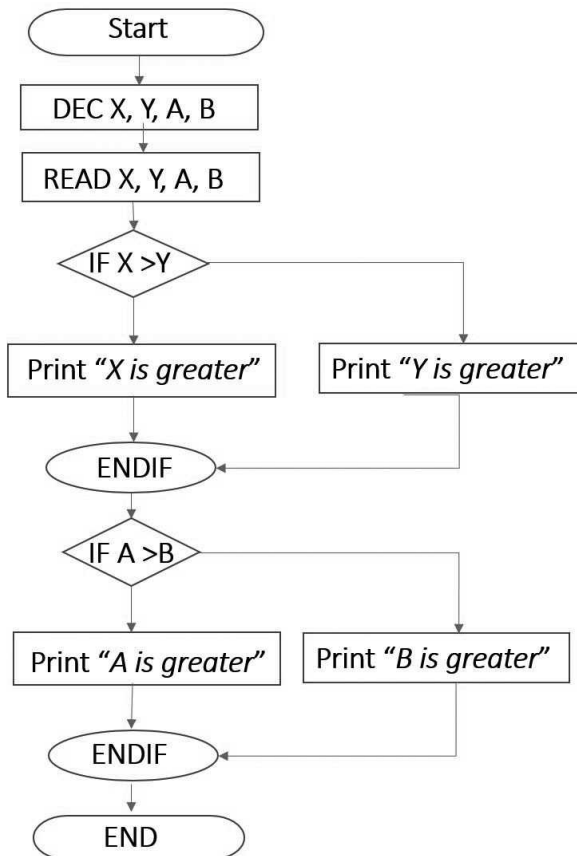
- a) Not correct – The number of tests for decision coverage does not change whether there is an executable statement for false outcome or not. See explanation below.
- b) Not correct – It is not necessary that the coverage for a particular test is dependent on the total number of decisions in the code. See explanation below.
- c) **Correct** – See explanation below.
- d) Not correct – It is possible. See explanation below.

Following code has two IF statements and both are independent of each other, with no other loops or case statements:

1. DEC X, Y, A, B integer
2. READ X, READ Y, READ A, READ B
3. IF X> Y THEN
4. “X is greater than Y”
5. ELSE
6. “Y is greater than X”
7. END IF
8. IF A> B THEN

9. "A is greater than B"
10. ELSE
11. "B is greater than A"
12. END IF

Flowchart for this code:



From the flow chart, it is clear that two tests are required to achieve 100% decision coverage and 100% statement coverage.

For the first test the values are $X=100$, $Y=50$, $A=100$, $B=50$

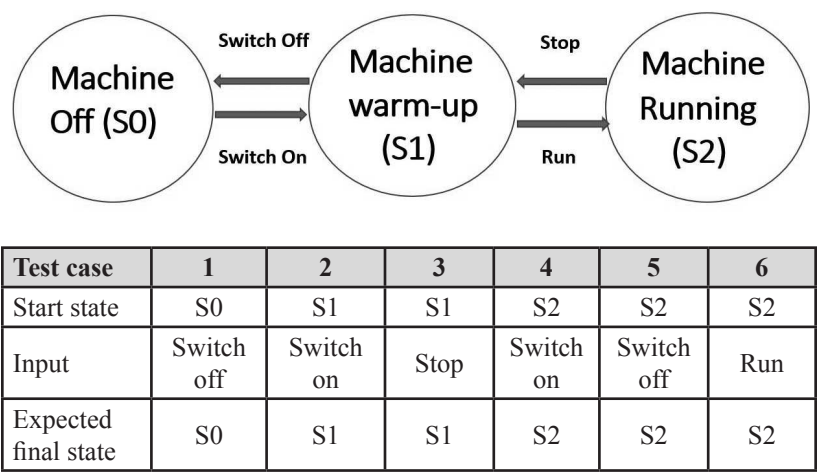
For the second test the values are $X=50$, $Y=100$, $A=50$, $B=100$

Therefore, in this case, two tests can provide 100% coverage in this case, also the number of tests will not change if there are no executable statements for IF statements.

Question 23

Consider a system for the treadmill machine used in the gym. The treadmill machine has to be in the warm-up state before it can be used for running. While the machine is in running state it cannot be switched-off.

The state transition diagram and table of test cases are shown below:



Which of the following statements are true about the given state transition diagram and table of test cases?

- a) The given test cases represent all the critical transitions in the state transition diagram.
- b) The given test cases can be used to cover some of the valid and invalid transitions in the state transition diagram.
- c) **The given test cases represent some of the invalid transitions in the state transition diagram. More tests are required to cover all the invalid transactions.**
- d) The given test cases represent all possible invalid transitions in the state transition diagram.

FL-4.2.4 (K3) Apply state transition testing to derive test cases from given requirements

Justification

A state table can be created for the above state transition diagram. All the states are on the left-hand side of the table, and all the triggers(events) are across the top.

Now to derive the value of each cell of the table input the trigger at the top of the current state.

Example: When the current state is **S0** and ‘**Switch on**’ trigger is applied the resulting state (end state) is **S1**. This is a valid transaction because when the trigger is applied the end state is different from the current state.

When the current state is **S1** and ‘**Switch on**’ trigger is applied nothing happens. Therefore, this is an invalid transaction or “null” transaction (denoted by ‘-’).

	Switch on	Switch off	Run	Stop
S0	S1	-	-	-
S1	-	S0	S2	-
S2	-	-	-	S1

From the table, there are 4 valid transactions and 8 invalid or “null” transactions (denoted by ‘-’).

The test case table represents only some of the invalid transactions but not all. Following additional two tests are required to cover all the invalid transactions.

Test case	7	8
Start state	S0	S0
Input	Run	Stop
Expected final state	S0	S0

- Not correct – It is not specified which transactions are critical.
- Not correct – Test cases do not cover any valid transactions.
- Correct** – See the explanation above.
- Not correct – Not all the invalid transactions are covered, see the explanation above.

Question 24

A leading mobile company is enhancing some of the features of an existing phone application. Existing business process diagrams defining the user interaction with the phone are updated based on the changes.

Which of the following test technique will be BEST suited for deriving test cases?

- a) State transition testing.
- b) Decision table testing.
- c) Checklist-based testing.
- d) **Use-case testing.**

FL-4.2.5 (K2) Explain how to derive test cases from a use case**Justification**

- a) Not correct – There is no information about the different states, therefore state transition testing is not possible.
- b) Not correct – There is no information regarding the decisions so decision table testing is not possible.
- c) Not correct – There is no information about any checklist which can be used for testing. Also, checklist-based testing requires experienced testers which is not stated in the question.
- d) **Correct** – As the business process diagrams are suitable to describe the interaction and activities, use-case testing will be best suited to derive test cases. (Syllabus 4.2.5)

Question 25

The bank ATM machine charges withdrawal fees for the other bank's card. The fee is calculated based on the amount withdrawn:

- For amount \$500 or less the fee is \$1.
- For amount over \$500 and up to \$1000, the fee is \$2.
- For any amount more than \$1000 the fee is \$3.

How many tests will be required to test all set of values (\$) for three-point boundary value analysis? The withdrawal amounts can only be in multiples of \$1.

- a) 10
- b) 9
- c) **8**
- d) 7

FL-4.2.2 (K3) Apply boundary value analysis to derive test cases from given requirements

Justification

The following partitions can be identified:

1. – 500 Three-point boundaries 499,500,501
2. 501 – 1000 Three -point boundaries 500, 501, 502, 999, 1000,1001
3. 1001 – Three -point boundaries 1000,1001,1002

Test are required for boundaries 499, 500, 501, 502, 999, 1000, 1001, 1002

Therefore, total tests required to test three-point boundaries are **8**.

Thus:

- a) Not correct – see the justification above.
- b) Not correct – see the justification above.
- c) **Correct** – see the justification above.
- d) Not correct – see the justification above.

Question 26

Which of the following statement regarding exploratory testing is TRUE?

- a) Exploratory testing requires some of the tests to be pre-defined and rest are designed dynamically during execution.
- b) Exploratory testing is not effective if the system is already tested thoroughly using formal testing techniques.
- c) Exploratory testing can provide higher coverage than other formal testing techniques if the testers are having strong experience in similar applications.
- d) **Exploratory testing is more effective when it is performed within a defined time-box using test charters.**

FL-4.4.2 (K2) Explain exploratory testing**Justification**

- a) Not correct – This is false. In exploratory testing, tests are not pre-defined but are designed, executed, logged, and evaluated dynamically during test execution. (Syllabus 4.4.2)
- b) Not correct – This is false. Exploratory testing can always be used to complement other more formal testing techniques. (Syllabus 4.4.2)
- c) Not correct – This is false. There is no guarantee that exploratory testing can provide higher coverage than other formal testing techniques. Also, testers strong experience in a similar application is not related to coverage.
- d) **Correct** – This is true. Test charter containing test objectives to guide the testing within the time-boxed session making exploratory testing more effective. (Syllabus 4.4.2)

Question 27

What is use case testing?

- a) A test technique in which tests are executed based on the tester's experience, knowledge and intuition.
- b) A test technique in which state table is used to derive test cases to validate valid and invalid transitions.
- c) A test technique in which test cases are derived based on the analysis of the internal structure of the system.
- d) **A test technique in which test cases are designed to execute scenarios of use cases.**

FL-4.x (K1) Keywords**Justification**

- a) Not correct – This is the definition of experience-based testing from the Glossary.
- b) Not correct – This is the definition of state transition testing from the Glossary.
- c) Not correct – This is the definition of white-box test technique from the Glossary.
- d) **Correct** – This is the definition of use case testing from the Glossary.

Question 28

While using the experience-based test technique for testing, which of the following TWO information would be MOST helpful?

- a) **Knowledge of the system environment.**
- b) knowledge of the internal structure of the system.
- c) **Knowledge of the expected use of the software.**
- d) Knowledge about the code of the test object.

FL-4.1.1 (K2) Explain the characteristics, commonalities, and differences between black-box test techniques, white-box test techniques and experience-based test techniques

Justification

- a) **Correct** – This will be relevant for experience-based test technique. (Syllabus 4.1.2)
- b) Not correct – This will be relevant for white box test technique. (Syllabus 4.1.2)
- c) **Correct** – This will be relevant for experience-based test technique. (Syllabus 4.1.2)
- d) Not correct – This will be relevant for white box test technique. (Syllabus 4.1.2)

Question 29

Consider a system built to automatically calculate the government stamp duty based on the purchase price of the house. There is no stamp duty for houses where the purchase price is less than \$200,000. The maximum stamp duty is 3.5% which is applied for houses where the purchase price is more than or equal to \$800,000.

Rest of the rules for calculating stamp duty are:

- 1.0% for houses with purchase price from \$200,000 and less than \$400,000
- 1.5% for houses with purchase price from \$400,000 and less than \$500,000
- 2.0% for houses with purchase price from \$500,000 and less than \$600,000
- 2.5% for houses with purchase price from \$600,000 and less than \$700,000
- 3.0% for houses with purchase price from \$700,000 and less than \$800,000

What is the minimum number of test cases required to cover all valid equivalence partitions for calculating the stamp duty?

- | | |
|------|------|
| a) 5 | c) 6 |
| b) 8 | d) 7 |

FL-4.2.1 (K3) Apply equivalence partitioning to derive test cases from given requirements

Justification

- a) Not correct – see explanation in d).
 b) Not correct – see explanation in d).
 c) Not correct – see explanation in d).
 d) **Correct** – Partitions are as below:
1. equivalence partition: $0 < \text{Cost of house} < \$200,000$
 2. equivalence partition: $\$200,000 < \text{Cost of house} < \$400,000$
 3. equivalence partition: $\$400,000 < \text{Cost of house} < \$500,000$
 4. equivalence partition: $\$500,000 < \text{Cost of house} < \$600,000$
 5. equivalence partition: $\$600,000 < \text{Cost of house} < \$700,000$
 6. equivalence partition: $\$700,000 < \text{Cost of house} < \$800,000$
 7. equivalence partition: $\$800,000 < \text{Cost of house}$.

Therefore, a minimum **seven** tests will be required to test all the valid partitions.

Question 30

For the following tests, organize the different degrees of test independence from the low level to high level.

- I. Tests designed by the testers from the same team.
- II. Tests designed by the test specialist from an independent test team.
- III. Tests designed by the developer who wrote the code.
- IV. Tests designed by another developer from the development team.
- V. Tests designed by the tester from the outsourced company.

- a) (III), (IV), (II), (I), (V)
- b) (III), (IV), (I), (V), (II)
- c) **(III), (IV), (I), (II), (V)**
- d) (V), (II), (I), (IV), (III)

FL-5.1.1 (K2) Explain the benefits and drawbacks of independent testing

Justification

Syllabus 5.1.1. A certain degree of independence (avoiding the author bias) often makes the tests more effective in finding defects. Levels of independence from low to high:

- Tests designed by the person(s) who created/coded the software under test (developer) **(III)**
- Tests designed by another person(s) (within the development team) **(IV)**
- Tests designed by testers (from the same team) **(I)**
- Tests designed by Independent test team specialist **(II)**
- Tests designed by testers from a different organization (outsourced company) **(V)**

Thus:

- a) Not correct
- b) Not correct
- c) **Correct**
- d) Not correct

Question 31

Which of the following statements regarding the entry and exit criteria for testing is FALSE?

- a) Exit criteria define what conditions must be achieved in order to declare a test level complete.
- b) Entry criteria define the preconditions for undertaking a given test activity.
- c) **Typical exit criteria include: Reliability measures, test cost, testable code ready, time to market, and product completeness.**
- d) Typical exit criteria include: Performance efficiency, residual risks and code coverage.

FL-5.2.3 (K2) Give examples of potential entry and exit criteria

Justification

- a) Not correct – This is true. (Syllabus 5.2.3)
- b) Not correct – This is true. (Syllabus 5.2.3)
- c) **Correct** – “Testable code ready” is typically a part of entry criteria for testing. (Syllabus 5.2.3)
- d) Not correct- This is true. (Syllabus 5.2.3)

Question 32

Which of the following statements correctly describe the tasks of the test manager and the tester?

- a) The test manager develops or reviews the test strategy to be followed, whereas the tester writes and updates the test plan.
- b) The test manager initiates the analysis, design, implementation, and execution of tests, whereas the tester prepares and delivers the test progress reports and test summary reports.
- c) The tester designs and implements test cases, whereas the test manager review tests developed by testers.
- d) **The test manager decides about the implementation of the test environment, whereas the tester designs, set up and verifies the test environment.**

FL-5.1.2 (K1) Identify the tasks of a test manager and tester**Justification**

- a) Not correct – It is the test manager’s task to write and update the test plan. (Syllabus 5.1.2)
- b) Not correct – It is the test manager’s task to prepares and deliver test progress reports and test summary reports. (Syllabus 5.1.2)
- c) Not correct – It is the testers task to review test cases prepared by other testers. (Syllabus 5.1.2)
- d) **Correct** – It is the test manager who decides about the implementation of the test environment, while the tester designs, set up and verifies the test environment. (Syllabus 5.1.2)

Question 33

Which of the following correctly describes the difference between metrics-based and expert-based techniques for test estimation?

- a) Metrics-based approach is applicable only for sequential development model, whereas expert-based approach is applicable for any development model.
- b) Metrics-based approach accuracy depends on the quantity of metrics, whereas expert-based approach accuracy depends on the years of experience of experts in their respective fields.
- c) **Metrics-based approach can be used if there is sufficient data from previous similar projects, whereas expert-based approach requires task experts for estimation.**
- d) Metrics-based approach can be used for all test level estimations, whereas expert-based approach is applicable for system level test estimations.

FL-5.2.6 (K2) Explain the difference between two estimation techniques: the metrics-based technique and the expert-based technique

Justification

- a) Not correct – Both can be used for any development model.
- b) Not correct – The accuracy of Metrics based approach accuracy depends not only on the quantity but the similarity of the project from where the data is collected. The expert-based approach is based on the expertise in the task not the years of experience.
- c) **Correct** – This correctly describes the metrics-based and expert-based techniques. (Syllabus 5.2.6)
- d) Not correct – Both can be used for any test levels.

Question 34

Given the following priorities and dependencies for the test cases created for a banking system:

Test ID	Test case Name	Dependency	Priority
TC1.	Login	None	Low
TC2.	Account Summary	TC1	High
TC3.	Transfer Payment	TC2	Medium
TC4.	Add Payee	TC3	High
TC5.	Check Statement	TC2	Medium
TC6.	Update Details	TC2	High

Which of the following test execution schedules BEST considers the priorities and dependencies?

- a) TC2 → TC1 → TC3 → TC4 → TC5 → TC6
- b) TC1 → TC2 → TC4 → TC6 → TC3 → TC5.
- c) TC1 → TC2 → TC6 → TC3 → TC5 → TC4.
- d) **TC1 → TC2 → TC6 → TC3 → TC4 → TC5.**

FL-5.2.4 (K3) Apply knowledge of prioritization, technical and logical dependencies to schedule test execution for a given set of test cases

Justification

- a) Not correct – Everything is dependent on TC1, so any test flow that does not start with TC1 is incorrect.
- b) Not correct – TC4 is dependent on TC3 so it should be tested before.
- c) Not correct – TC4 is having higher priority than TC5 so it should be tested before.
- d) **Correct** – Syllabus 5.2.4: Test cases are run based on their priority levels, usually by executing the test cases with the highest priority first. However, this practice may not work if the test cases have dependencies or the features being tested have dependencies.

In this scenarios the three highest priority test cases are TC2, TC4 and TC6. TC4 is dependent on TC3 which in turn is dependent on TC2,

TC4 is dependent on TC2. That means TC2 needs to be executed first but as it dependent on TC1 therefore the first two test cases should be scheduled as TC1 → TC2.

Now TC4 & TC6 are having same priority but TC4 is dependent on TC3 which is having medium priority so TC6 will be executed first followed by TC3 and TC4. TC5 will be executed in the last.

So, the sequence of test execution taking the dependencies and priorities into account will be

TC1 → TC2 → TC6 → TC3 → TC4 → TC5.

Question 35

Which of the following metrics would be MOST useful to measure progress during the test execution?

- a) **Percentage of test conditions passed/failed.**
- b) Percentage of planned test cases implemented.
- c) Details of defects closed during test execution.
- d) Resource allocation and usage during test execution.

FL-5.3.1 (K1) Recall metrics used for testing

Justification

- a) **Correct** – Percentage of test conditions passed/failed would be helpful to assess progress during the test execution. (Syllabus 5.3.1)
- b) Not correct – Percentage of planned test cases implemented would be helpful to assess progress against the test preparation. (Syllabus 5.3.1)
- c) Not correct – Details of defects closed during test execution will be helpful to assess the current quality of the test object. (Syllabus 5.3.1)
- d) Not correct – Resource allocation and usage during execution will be helpful to assess the progress against the budget. (Syllabus 5.3.1)

Question 36

Which of the following should be included as a part of the test progress report?

- I. The status of the test activities and progress against the test plan.
- II. Factors impeding the test progress.
- III. Lessons learned from the completed test activities.
- IV. Testing activities planned for the next reporting period.
- V. The current quality of the test object.

- a) I, II
- b) I, II, IV
- c) **I, II, IV, V**
- d) All of the above

FL-5.3.2 (K2) Summarize the purposes, content, and audiences for test reports

Justification

- I. This is important for the test status report. (Syllabus 5.3.2)
- II. This is important for the test status report. (Syllabus 5.3.2)
- III. This is not important for the test status report. This may be required during test completion activities.
- IV. This is important for the test status report. (Syllabus 5.3.2)
- V. This is important for the test status report. (Syllabus 5.3.2)

Thus:

- a) Not correct
- b) Not correct
- c) **Correct**
- d) Not correct

Question 37

Which of the following statement correctly describes the difference between analytical and reactive test strategy for testing?

- a) Analytical test strategy is based on the analysis of risk, whereas reactive test strategy is based on the instructions of the business domain experts.
- b) Analytical test strategy implements tests based on external rules and standards, whereas in reactive test strategy tests are implemented based on the prior test results.
- c) **In analytical test strategy, tests and their order are pre-planned based on risk analysis, whereas in reactive test strategy tests are not pre-planned but are created based on the prior tests results.**
- d) Analytical test strategy uses extensive automated regression tests, whereas reactive test strategy uses extensive exploratory tests.

FL-5.2.2 (K2) Differentiate between various test strategies

Justification

- a) Not correct – This describes the consultative, not the reactive test strategy. (Syllabus 5.2.2)
- b) Not correct – This describes the process-compliant, not the analytical test strategy. (Syllabus 5.2.2)
- c) **Correct** – Analytical test strategy uses risk analysis to plan the tests and their order while in reactive test strategy tests are not pre-planned but created based on the previous test results test. (Syllabus 5.2.2)
- d) Not correct – It is the regression-averse test strategy which uses extensive automated regression tests and not the analytical test strategy. (Syllabus 5.2.2)

Question 38

A new tester in the project is responsible for testing the front end of the system which will be used by the internal staff. There are two test users which can be used for testing the internal staff functionality.

While testing the user detail page, the tester has observed that static text for all the input fields is starting with lowercase letters. For both the test users the observation is similar.

In the specification document, it was clearly mentioned that all the static fields should start with an uppercase letter.

Tester has written the defect report to highlight this issue with the following information:

Defect ID: 1009

Title: Static text starts with lowercase in user detail page

Short summary: After the internal user login to the front-end application, the user detail page shows all the static fields in lowercase letters.

Expected result: All the static fields should start with an uppercase letter.

What is the MOST important piece of information that is missed by the tester in the above defect report?

- a) Test data used for testing.
- b) The actual test results.
- c) **Severity and priority of the defect.**
- d) Date and author of the defect

FL-5.6.1 (K3) Write a defect report, covering defects found during testing.

Justification

- a) Not correct – This is not important as the behavior is similar for all the users.
- b) Not correct – This is not important as the details are already in the summary.
- c) **Correct** – This is a piece of important information which is required from the tester. Looking closely at the situation, as it is an internal application, this issue is having low severity and low priority to fix. This need to be communicated to the developers to help them decide which defect to fix first. (Syllabus 5.6)
- d) Not correct – This information is missing but is not so important for this situation.

Question 39

Which of the following should be BEST considered while successfully implementing a new tool within an organization?

- a) The organization should not plan to start any test activity unless the complete rollout for new tool is completed.
- b) The organization should plan for complete rollout of the tool in one go rather than incrementally.
- c) The organization should plan to make changes to its development lifecycle to fit with the use of the tool.
- d) **The organization should plan that processes should be adapted and improved to fit with the use of the tool.**

FL-6.2.3 (K1) Identify the success factors for evaluation, implementation, deployment, and on-going support of test tools in an organization

Justification

- a) Not correct – New testing activities are not dependent on the complete rollout and will not affect the successful implementation of the tool.
- b) Not correct – Tool should be rolled out incrementally to minimize the risk. The lessons learned from the initial rollouts can be helpful for the later rollouts.
- c) Not correct – Organization development lifecycle is not changed to fit with the use of the tool.
- d) **Correct** – Adapting and improving the process to fit with the use of the tool will help to get maximum benefit from the tool. (Syllabus 6.2.3)

Question 40

Which of the following is a characteristic of a test harness tool?

- a) It helps to enable tests for automatic/semi-automatic execution using stored inputs.
- b) It helps to store information about versions and builds of software and testware.
- c) **It helps to provide a predictable and controllable environment in which the faults can be localized.**
- d) It helps to find inconsistencies in the database model.

FL-6.1.1 (K2) Classify test tools according to their purpose and the test activities they support

Justification

- a) Not correct – This is the characteristic of a test execution tool.
- b) Not correct – This is the characteristic of configuration management tool.
- c) **Correct** – This is the characteristic of Test harness tool.
- d) Not correct – This is the characteristic of model based testing tool.

6

Practice Exam 3

Question 1

Which of the following statements is TRUE?

- I. Programming error can lead to a defect in the code.
- II. All unexpected test results can lead to failures.
- III. A defect in the code will definitely cause a failure.
- IV. False positives are reported as defects, but they may not be defects.

- a) I and III
- b) II and III
- c) I and IV
- d) II and IV

Select one option.

Question 2

Which of the following are examples of Test execution work products?

- I. Defect reports.
- II. Test summary reports.
- III. Test suites.
- IV. Test execution schedule.

- a) I, II & III.
- b) II, III & IV.
- c) I & IV.
- d) only I.

Select one option.

Question 3

Which of the following statements correctly describes the confirmation test?

- a) A test activity which checks the source of defects and confirms fixes.
- b) A test activity which confirms that defects are resolved by the fixes.
- c) A test activity which confirms the defect status before the fix is applied.
- d) A test activity which removes faults after the debugging is completed.

Select one option.

Question 4

Which of the general testing principles (1-4) are characterized by the description(A-D) below?

- 1. Defect clustering
 - 2. Pesticide Paradox
 - 3. Absence of error fallacy
 - 4. Testing shows presence of defects
-
- A. If all the defects are found and fixed, the system can still be unusable.
 - B. If no defects are found during testing, it is not a proof of correctness.
 - C. A small number of modules are responsible for most of the operational failures.
 - D. Test cases should be reviewed and revised regularly to find new defects.
-
- a) 1D, 2C, 3B, 4A.
 - b) 1C, 2D, 3B, 4A
 - c) 1C, 2D, 3A, 4B
 - d) 1A, 2C, 3D, 4B.

Select one option.

Question 5

Which of the following BEST describes the advantages of having measurable coverage criteria defined for the test basis?

- a) It helps in defining the objective of testing and approach for meeting this objective.
- b) It helps in the test design activity to effectively mitigate product risks.
- c) It helps to inform the stakeholders whether specified requirements are fulfilled after the test execution.
- d) It helps to follow the standard ISTQB test process which is required for all standards projects.

Select one option.

Question 6

Which one of the following is the BEST description of a test suite?

- a) A sequence of test cases in order of execution order which requires set up of initial preconditions and any wrap-up activities post-execution.
- b) A set of test scripts or test procedures to be executed in a specific test run.
- c) Documentation specifying the features to be tested and their corresponding test conditions.
- d) Documentation describing the test objectives and the means and schedule to achieve them, that is organized to coordinate testing activities.

Select one option.

Question 7

Which of the following will **LEAST** affect the objective of testing for any given project?

- a) The context of the system being tested.
- b) Stakeholders of the project.
- c) Software development lifecycle model.
- d) Test level.

Select one option.

Question 8

Map the following test tasks (1-4) with their respective ISTQB Test process activity (A-D)

1. Creating test suites for testing.
2. Capturing bi-directional traceability between the test basis, test conditions, and test cases.
3. Creating a test summary report for the stakeholders.
4. Assessing the quality of a system based on test results.

- A. Test implementation
- B. Test monitoring and control
- C. Test design
- D. Test completion

- a) 1A, 2D, 3B, 4C.
- b) 1D, 2B, 3A, 4C.
- c) 1B, 2A, 3D, 4A.
- d) 1A, 2C, 3D, 4B.

Select one option.

Question 9

For maintenance testing in the project, which of the following conditions can MOST likely make the impact analysis task difficult?

- I. Current test data in the test environment is out of date.
- II. Business requirements are missing.
- III. Test environment of the project is currently unavailable.
- IV. The people involved in impact analysis are new to the project.

- a) I and II
- b) III and IV
- c) II and IV
- d) II and III

Select one option.

Question 10

Which of the following statements (1-4) are related to different test type (A-D)?

1. Test design and execution requires knowledge of changes made to the system.
 2. Test design and execution requires knowledge of how the code is built and how the data is stored.
 3. Test design and execution requires knowledge of the business domain of the system.
 4. Test design and execution requires knowledge of the inherent drawbacks of design or technology.
- A. Functional testing.
 - B. Non-functional testing.
 - C. White-box testing.
 - D. Change-related testing.
- a) 1D, 2B, 3A, 4C
 - b) 1D, 2C, 3A, 4B
 - c) 1C, 2B, 3A, 4D
 - d) 1D, 2A, 3B, 4C

Select one option.

Question 11

Which of the following statements is a characteristic of good testing and applies to any software development life cycle model?

- I. All the test levels are completed for each developed feature.
 - II. Each test level in the development lifecycle has a test objective specific to that level.
 - III. Test analysis and design for a given test level begins after the corresponding development activity is completed.
 - IV. Testers are involved in discussion and review of all the work products as soon as the drafts are available.
- a) I & II
 - b) III & IV
 - c) II & IV
 - d) II & III

Select one option.

Question 12

How can white box tests be applied during component testing?

- a) By verifying that all work process flows are covered.
- b) By verifying that statement coverage and code decision coverage for all paths in the code is achieved.
- c) By verifying that all web page navigations are covered.
- d) By verifying that large volumes of data can be successfully transferred between components.

Select one option.

Question 13

Which of the following work products (1-4) are commonly used as a test basis for the different level of testing (A-D)?

- 1. Installation procedures
 - 2. Data models
 - 3. Workflows
 - 4. State diagrams
-
- A. Component testing
 - B. Integration testing
 - C. System testing
 - D. Acceptance testing
-
- a) 1D, 2A, 3B, 4C.
 - b) 1D, 2B, 3A, 4C.
 - c) 1C, 2D, 3B, 4A.
 - d) 1A, 2C, 3D, 4B.

Select one option.

Question 14

For the following situation (1-4) below, choose the BEST suited review types (A-D)?

1. Review of the Test plan document.
2. Review of the design document for a safety critical system.
3. Review of peer test cases.
4. Review of Technical work products for a complex system.

- A. Technical Review.
- B. Informal Review.
- C. Inspection.
- D. Walkthrough.

- a) 1D, 2C, 3B, 4A.
- b) 1B, 2C, 3D, 4A.
- c) 1D, 2B, 3C, 4A.
- d) 1B, 2D, 3C, 4A.

Select one option.

Question 15

A mobile company is developing software for its new series of phones. The detailed specifications are captured in the form of use-cases. These use cases have detailed descriptions of how mobile phone users use the mobile phone for the following flow of events:

- Making phone call
- Receiving phone call
- Viewing call history
- Sending SMS
- Maintaining contacts

The team is using perspective-based review techniques to review these use-cases. For the reviewer who has taken the role of “tester”, which of the following will be appropriate to check the effectiveness of the work-product?

- a) Try to find the defects in the work product from a tester’s perspective.
- b) Find the issues in the work product based on the potential defects list.
- c) Generate a draft of end-to-end test cases using the work product.
- d) Review the work product sequentially, identifying and documenting the issues as they are encountered.

Select one option.

Question 16

Which of the following TWO statements about static testing are true?

- a) Static testing is an important part of security testing.
- b) Static testing decreases the development cost but increases the testing cost.
- c) Static testing increases development productivity.
- d) Static testing can find run-time issues before dynamic testing.

Select two options.

Question 17

In the formal review, who is primarily responsible for collating and recording potential defects during the review activity?

- a) Facilitator
- b) Reviewers
- c) Author
- d) Scribe

Select one option.

Question 18

Which of the following activities are related to “Initiate review” activity of the work product review process?

- I. Selecting the participants and allocating roles.
 - II. Distributing the work product and supporting material to the participants.
 - III. Estimate effort and timeframe.
 - IV. Review of the work product.
-
- a) I and II
 - b) II and III
 - c) II, III and IV
 - d) Only II

Select one option.

Question 19

A car park system calculates the parking fee based on the total time the car is in the parking area. The total time is calculated as the difference between the entry and exit time.

- If the total time is less than or equal to 8 hours the fee is \$11
- For total time more than 8 hours up to 20 hours the fee is \$28
- After that, for each 2-hour slot, \$5 is added to the fee.

The maximum time for which the passenger can be charged is 24 hours and time is measured to the nearest minute.

Which of the following are a valid set of values (in hrs.mins) identified by two-point boundary value analysis?

- a) 0, 1, 8, 9, 19, 20, 21, 22, 23, 24
- b) 0.0, 0.1, 8.01, 20.00, 20.01, 22.00, 23.59
- c) 7.59, 8.00, 8.01, 19.59, 20.00, 20.01, 22.00, 22.01
- d) 8.00, 8.01, 20.00, 20.01, 22.00, 22.01, 24.00, 24.01

Select one option.

Question 20

Consider the following statement:

“The code contains only a single IF statement with an explicit true statement and no explicit false statement.”

Which of the following is correct with respect to the above scenario?

- a) For this scenario, the number of tests required for 100% statement coverage should be equal to the tests required for 100% decision coverage.
- b) For this scenario, the number of tests required for 100% statement coverage should be less than the tests required for 100% decision coverage.
- c) For this scenario, a number of tests required for 100% statement coverage should be more than the tests required for 100% decision coverage.
- d) For this scenario, additional information is required to get the exact number of tests to achieve 100% statement and 100% decision coverage.

Select one option.

Question 21

Which one of the following is the BEST description of decision coverage?

- a) It is measured as all the true decision outcomes in the source code divided by the total number of decision outcomes in the code expressed as a percentage.
- b) It is measured as the number of decision outcomes executed by the tests divided by the total number of decision outcomes in the code, expressed as a percentage.
- c) It is measured as the number of decision outcomes of the source code executed by test cases that are passed, expressed as a percentage.
- d) It is measured as the number of decision outcomes divided by the statement coverage, expressed as a percentage.

Select one option.

Question 22

Which of the following statements describing statement coverage and decision coverage are true?

- I. 100% statement coverage does not ensure that all decisions are tested whereas 100% decision coverage ensures all statements are tested.
 - II. Statement testing may provide less coverage than decision testing.
 - III. Statement coverage finds defects in code that was not exercised by other tests and decision coverage finds defects in the code where other tests have not taken into account the false outcomes.
 - IV. Statement testing and decision testing will provide the same coverage when testing a code having an IF statement without an ELSE statement.
-
- a) I and II.
 - b) I, II and III.
 - c) I, II and IV.
 - d) I, II, III and IV.

Select one option.

Question 23

Which of the following BEST defines the coverage?

- The degree to which a component or system can be used by specified users.
- The degree to which specified coverage items have been exercised by a test suite expressed as a percentage.
- The degree of effectiveness and efficiency with which tests can be designed and executed for a system.
- The degree to which a relationship can be established between two or more work products.

Select one option.

Question 24

A new online system is developed to generate a weekly train pass for different categories of commuters. To buy a weekly pass, the user must select only one of the option from the following category:

- Office staff
- Students
- Foreign tourists

The following decision table has been designed to test the logic and decide whether the user qualifies for the discount.

	R1	R2	R3	R4	R5	R6	R7	R8
Conditions								
Office staff	Y	N	Y	N	Y	N	Y	N
Students	N	N	Y	Y	N	N	Y	Y
Foreign tourists	N	N	N	N	Y	Y	Y	Y
Actions								
Discount	N	N	N	Y	N	Y	N	N

Which of the test cases wouldn't occur in a real situation and can be eliminated from the decision table?

- R1, R2, R3, R5, R7, and R8.
- R2.
- R2, R3, R5, R7, and R8.
- R3, R5, R7 and R8.

Select one option.

Question 25

Which of the following can be used for creating test conditions, test cases, and test data for white-box testing techniques?

- I. Business Processes
 - II. Distribution of likely defects
 - III. Detailed design of the system
 - IV. Code of the test object
- a) I & II.
 - b) III & IV.
 - c) I & III.
 - d) II & IV.

Select one option.

Question 26

An online train reservation system calculates ticket concession based on the passenger's age. The age can only be entered as a positive number, without any decimal or fractional parts.

- For passenger aged, 5 or less the concession is 100%.
- For passenger aged over 5 and up to 18, the concession is 60%.
- For passenger aged over 18 and up to 60, there is no concession.
- For a passenger aged more than 60, the concession is 50%.

Which of the following is the most likely set of values (in years) identified by two-point boundary value analysis?

- a) 4, 5, 6, 17, 18, 19, 59, 60, 61
- b) 5, 18, 60
- c) -1, 0, 5, 6, 18, 19, 60
- d) 5, 6, 18, 19, 60, 61

Select one option.

Question 27

Which of the following TWO statements clearly describe the difference between exploratory testing and error guessing?

- a) Exploratory testing is applicable only for system level, whereas error guessing is applicable for all test levels.
- b) In Exploratory testing, tests are not pre-defined, whereas in error guessing the tests are pre-defined.
- c) In Exploratory testing test results are used to create additional tests for areas that need more testing, whereas in error guessing previous defect list can be used to create additional tests for areas that need more testing.
- d) Exploratory testing requires testers to be experienced in similar applications, whereas for error guessing technique testers experience is not important.

Select two options.

Question 28

The system for income tax department calculates the tax based on the individual income earned during the financial year. Income up to \$20,000 is exempted from tax, thereafter tax is calculated based on the following rules:

- For income up to \$45,000 the tax is \$5672 plus 4% of the amount above \$20,000
- For income more than \$45,000 and up to \$75,000 the tax is \$6670 plus 6% of the amount above \$45,000
- For income more than \$75,000 and up to \$120,000 the tax is \$9628 plus 8% of the amount above \$75,000
- For income more than \$120,000 the tax is \$19,672 plus 10% of the amount above \$120,000

Which of these groups of numbers which cover all the valid equivalence classes?

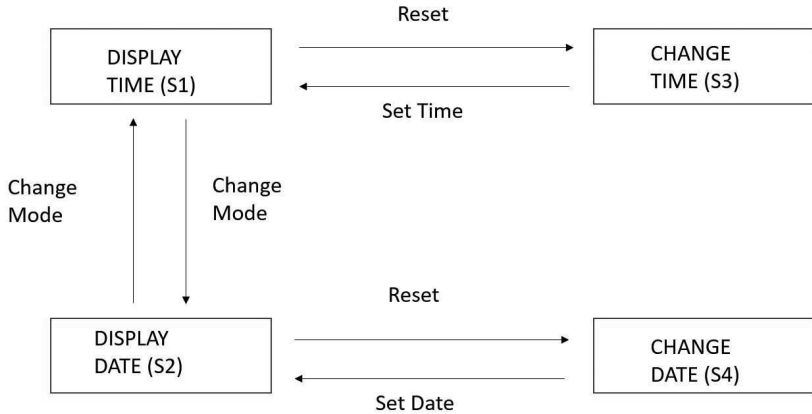
- a) (\$0), (\$20,000), (\$45,000), (\$75,000), (\$120,000)
- b) (\$20,000), (\$45,000), (\$75,000), (\$120,000), (\$150,000)
- c) (\$20,000), (\$45,000), (\$75,000), (\$120,000)
- d) (\$45,000), (\$75,000), (\$120,000), (\$150,000)

Select one option.

Question 29

Consider a system for an electronic device which can display either time or the date based on the user selection.

Following state transition diagram shows the different states and input requests for this device.



How many tests are required to test all the invalid transactions for the above state transition diagram?

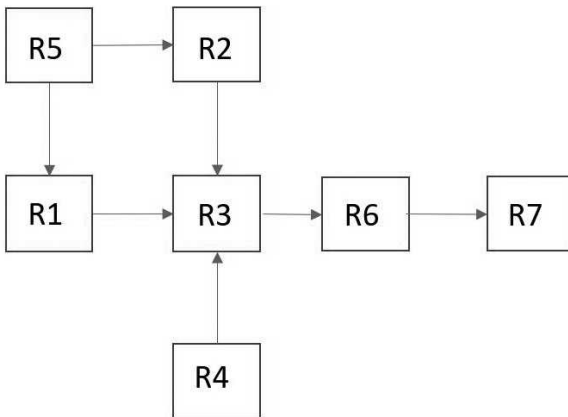
- a) 16 tests.
- b) 6 tests.
- c) 9 tests.
- d) 10 tests.

Select one option.

Question 30

The following diagram shows the logical dependencies between a set of seven requirements, where a dependency is shown by an arrow.

R2 and R3 are having the highest priority for execution followed by R6. The remaining requirements are having low priority.



If requirement with the highest priority is executed first, which one of the following options structures the test execution schedule?

- a) R2 → R3 → R6 → R5 → R1 → R4 → R7.
- b) R2 → R3 → R5 → R1 → R4 → R3 → R6 → R7.
- c) R5 → R2 → R1 → R4 → R3 → R6 → R7.
- d) R5 → R1 → R2 → R4 → R3 → R6 → R7.

Select one option.

Question 31

You are testing the software of an electronic watch. This watch can display the current time of different countries based on the selection. During testing, it was observed that for few countries the displayed current time is not accurate.

You have written the following defect report with the following information.

Title: Displayed current time not accurate

Summary: For the country codes (XYZ,...,..) the current time is not displayed correctly in the watch. For the rest of the country codes, the current time is correct.

Conclusion: This may be due to the daylight-saving logic not working correctly.

Which of the following statement about the testers “conclusion” on this defect is correct?

- a) This information is not important as it may confuse the developer who is fixing this problem.
- b) This information is not required as it is the developer’s job to find the root cause of defects.
- c) This information is important as it may provide additional information required for fixing this problem.
- d) This information is not part of the standard defect report so does not provide any value in the defect report.

Select one option.

Question 32

Which of the following are potential drawbacks of independent testing?

1. Independent testers will increase the overall project cost.
 2. Independent testers can be biased towards the developers.
 3. Independent testers may be seen as a bottleneck in the project release.
 4. Independent testers may not have all the information needed to perform testing.
- a) 1 and 2.
 - b) 1 and 3.
 - c) 3 and 4.
 - d) 2 and 4.

Select one option.

Question 33

Which of the following TWO are included as a part of the test summary report?

- a) Defining the scope and objectives of testing.
- b) Deviations from the test schedule.
- c) Budgeting of the test activities.
- d) Reusable test work products produced.

Select two options.

Question 34

Which of the following is the **LEAST** likely one to influence the test effort estimation for a particular project?

- a) The suspension and resumption criteria for testing.
- b) The complexity of the product domain.
- c) The tools used for testing.
- d) The type and quality of test basis.

Select one option.

Question 35

Which of the following is **NOT** the objective of product risk analysis?

- a) Determine additional activities other than testing to reduce risk.
- b) Determine the test techniques to be used.
- c) Determine the different levels of testing required.
- d) Determine the extent of independent testing required.

Select one option.

Question 36

Which of the following is the key task for the test manager?

- a) Document deviations from expected results.
- b) Check the status of exit criteria.
- c) Use appropriate tools to facilitate test process.
- d) Coordinate with network and other teams for environment setup.

Select one option.

Question 37

Which of the following factors are **MOST** likely to be considered while tailoring the test strategy for a project?

- I. Complexity of the project
 - II. Product risk analysis
 - III. Test estimation technique used
 - IV. Goals of the project
- a) I, II, III
 - b) I, III, IV
 - c) II, III, IV
 - d) I, II, IV

Select one option.

Question 38

Which of the following TWO can be considered as the objectives of a good configuration management system to support testing activities?

- a) To ensure a relationship is established between each test items.
- b) To ensure proper utilization of tools for all test activities.
- c) To ensure software items are referenced unambiguously in test documentation.
- d) To ensure all the testing activities follow the test process.

Select two options.

Question 39

Which of the following is MOST likely to be considered when selecting a new tool for an organization?

- a) Release cycle of the organization.
- b) The current quality of the test object.
- c) Training required for the users of new tool.
- d) The defect density of the projects where the tool can be used.

Select one option.

Question 40

Which of the following statements related to the purpose of testing tools is True?

- I. Testing tools increase the reliability of testing.
 - II. Testing tools can eliminate the need for having some test levels.
 - III. Testing tools have a higher level of defect reproducibility.
 - IV. Testing tools can automate activities that cannot be done manually.
- a) I, II and III
 - b) I, III and IV
 - c) I and IV
 - d) I, II, III and IV

Select one option.

Answers

1	C	K2
2	D	K2
3	B	K2
4	C	K2
5	C	K2
6	B	K1
7	B	K1
8	D	K2
9	C	K2
10	B	K2
11	C	K2
12	B	K1
13	A	K2
14	A	K2
15	C	K3
16	A, C	K2
17	D	K1
18	D	K2
19	D	K3
20	B	K2

21	B	K2
22	A	K2
23	B	K1
24	C	K3
25	B	K2
26	D	K3
27	B, C	K2
28	B	K3
29	D	K3
30	C	K3
31	C	K3
32	C	K2
33	B, D	K2
34	A	K1
35	D	K2
36	B	K1
37	D	K2
38	A, C	K2
39	C	K1
40	B	K2

7

Answers – Practice Exam 3

Question 1

Which of the following statements is TRUE?

- I. Programming error can lead to a defect in the code.
- II. All unexpected test results can lead to failures.
- III. A defect in the code will definitely cause a failure.
- IV. False positives are reported as defects, but they may not be defects.

- a) I and III
- b) II and III
- c) **I and IV**
- d) II and IV

FL-1.2.3 (K2) Distinguish between error, defect, and failure

Justification

- I. This is **true**. An error by the programmer due to requirement elicitation error or misinterpretation of requirement can lead to a defect in the code. (Syllabus 1.2.3)
- II. This is **false**. Not all unexpected test results are failures. It may happen that these unexpected test results are occurring because the tests were executed in a wrong way, or due to defects in the test data, the test environment, other testware, etc. (Syllabus 1.2.3)
- III. This is **false**. It is not necessary that all defects will cause failure as some defects require very specific inputs or preconditions to trigger a failure, which may occur rarely or never. (Syllabus 1.2.3)
- IV. This is **true**. False positive may be due to reason the tests were executed in a wrong way, or due to defects in the test data, the test environment, other testware, etc. (Syllabus 1.2.3)

Thus:

- a) Not correct
- b) Not correct
- c) **Correct**
- d) Not correct

Question 2

Which of the following are examples of Test execution work products?

- I. Defect reports.
- II. Test summary reports.
- III. Test suites.
- IV. Test execution schedule.

- a) I, II & III.
- b) II, III & IV.
- c) I & IV.
- d) **only I.**

FL-1.4.3 (K2) Differentiate the work products that support the test process

Justification

- I. **Defect reports** are Test execution work product. Syllabus 1.4.3.
- II. **Test summary report** is Test completion work product. Syllabus 1.4.3.
- III. **Test suites** are Test implementation work product. Syllabus 1.4.3.
- IV. **Test execution schedule** is a Test implementation work product. Syllabus 1.4.3.

Thus:

- a) Not correct
- b) Not correct
- c) Not correct
- d) **Correct**

Question 3

Which of the following statements correctly describes the confirmation test?

- a) A test activity which checks the source of defects and confirms fixes.
- b) A test activity which confirms that defects are resolved by the fixes.**
- c) A test activity which confirms the defect status before the fix is applied.
- d) A test activity which removes faults after the debugging is completed.

FL-1.1.2 (K2) Differentiate testing from debugging**Justification**

- a) Not correct. Confirmation test does not check the source of defects.
- b) Correct.** Confirmation test confirms whether the fixes have resolved the defects. (Syllabus 1.1.2)
- c) Not correct. Confirmation test is performed after the fix is applied and not before.
- d) Not correct. Confirmation testing does not remove faults.

Question 4

Which of the general testing principles (1-4) are characterized by the description(A-D) below?

1. Defect clustering
 2. Pesticide Paradox
 3. Absence of error fallacy
 4. Testing shows presence of defects
-
- A. If all the defects are found and fixed, the system can still be unusable.
 - B. If no defects are found during testing, it is not a proof of correctness.
 - C. A small number of modules are responsible for most of the operational failures.
 - D. Test cases should be reviewed and revised regularly to find new defects.
-
- a) 1D, 2C, 3B, 4A.
 - b) 1C, 2D, 3B, 4A
 - c) **1C, 2D, 3A, 4B**
 - d) 1A, 2C, 3D, 4B.

FL-1.3.1 (K2) Explain the seven testing principles

Justification

- Principle#4. **Defect clustering-** A small number of modules usually contain most of the defects discovered during pre-release testing, or is responsible for most of the operational failures. (Syllabus 1.3) **(1C)**
- Principle#5. **Pesticide paradox-** If the same tests are repeated over and over again, eventually these tests no longer find any new defects. To detect new defects, existing tests and test data may need changing, and new tests may need to be written. (Syllabus 1.3) **(2D)**
- Principle#7. **Absence of error fallacy-** It is a fallacy to expect that just finding and fixing a large number of defects will ensure the success of the system. It is possible that the system still does not fulfill the users' needs and expectations and is unusable. (Syllabus 1.3) **(3D)**
- Principle#1. **Testing shows presence of defects-** Testing can show that defects are present, but cannot prove that there are no

defects. Testing reduces the probability of undiscovered defects remaining in the software but, even if no defects are found, testing is not a proof of correctness. (Syllabus 1.3) **(4B)**

Thus:

- a) Not correct
- b) Not correct
- c) **Correct**
- d) Not correct

Question 5

Which of the following BEST describes the advantages of having measurable coverage criteria defined for the test basis?

- a) It helps in defining the objective of testing and approach for meeting this objective.
- b) It helps in the test design activity to effectively mitigate product risks.
- c) **It helps to inform the stakeholders whether specified requirements are fulfilled after the test execution.**
- d) It helps to follow the standard ISTQB test process which is required for all standards projects.

FL-1.4.1 (K2) Explain the impact of context on the test process

Justification

- a) Not correct- This is a test planning activity which is not related to measurable coverage criteria.
- b) Not correct- The test design activity is not related to measurable coverage criteria.
- c) **Correct-** The coverage criteria can act effectively as key performance indicators (KPIs) to drive the activities that demonstrate achievement of software test objectives. If all the elements of test basis are covered by a test case, the results of these tests after execution can tell the stakeholders whether specified criteria were fulfilled. (Syllabus 1.4.1)
- d) Not correct- This statement is not valid.

Question 6

Which one of the following is the BEST description of a test suite?

- a) A sequence of test cases in order of execution order which requires set up of initial preconditions and any wrap-up activities post-execution.
- b) **A set of test scripts or test procedures to be executed in a specific test run.**
- c) Documentation specifying the features to be tested and their corresponding test conditions.
- d) Documentation describing the test objectives and the means and schedule to achieve them, that is organized to coordinate testing activities.

FL-1.x (K1) Keywords Chapter 1**Justification**

- a) Not correct – This is the definition of test procedure according to Glossary.
- b) **Correct** – From Glossary.
- c) Not correct – This is the definition of test design specification according to Glossary
- d) Not correct – This is the definition of test plan according to Glossary.

Question 7

Which of the following will **LEAST** affect the objective of testing for any given project?

- a) The context of the system being tested.
- b) Stakeholders of the project.**
- c) Software development lifecycle model.
- d) Test level.

FL-1.1.1 (K1) Identify typical objectives of testing**Justification**

- a) Not correct – This may widely affect the objective of testing. (Syllabus 1.1.1)
- b) Correct** – Project stakeholders will least affect the objective of testing.
- c) Not correct – This may widely affect the objective of testing. (Syllabus 1.1.1)
- d) Not correct – This may widely affect the objective of testing. (Syllabus 1.1.1)

Question 8

Map the following test tasks (1-4) with their respective ISTQB Test process activity (A-D)

1. Creating test suites for testing.
 2. Capturing bi-directional traceability between the test basis, test conditions, and test cases.
 3. Creating a test summary report for the stakeholders.
 4. Assessing the quality of a system based on test results.
-
- A. Test implementation
 - B. Test monitoring and control
 - C. Test design
 - D. Test completion
-
- a) 1A, 2D, 3B, 4C.
 - b) 1D, 2B, 3A, 4C.
 - c) 1B, 2A, 3D, 4A.
 - d) 1A, 2C, 3D, 4B.

FL-1.4.2 (K2) Describe the test activities and respective tasks within the test process

Justification

- Creating test suites for the testing is done during **Test implementation** activity. Syllabus 1.4.2. **(1A)**
- Capturing bi-directional traceability between the test basis, test conditions and test cases are done during **Test design** activity. Syllabus 1.4.2. **(2C)**
- Creating a test summary report for the stakeholders is done during **Test completion** activity. Syllabus 1.4.2. **(3D)**
- Assessing the quality of a system based on test results is done during **Test monitoring and control** activity. Syllabus 1.4.2. **(4B)**

Thus:

- a) Not correct
- b) Not correct
- c) Not correct
- d) **Correct**

Question 9

For maintenance testing in the project, which of the following conditions can MOST likely make the impact analysis task difficult?

- I. Current test data in the test environment is out of date.
- II. Business requirements are missing.
- III. Test environment of the project is currently unavailable.
- IV. The people involved in impact analysis are new to the project.

- a) I and II
- b) III and IV
- c) **II and IV**
- d) II and III

FL-2.4.2 (K2) Describe the role of impact analysis in maintenance testing

Justification

- I. This can make the actual maintenance testing difficult not the impact analysis task.
- II. If the business requirements are missing or out of date it can make the impact analysis task difficult. Syllabus 2.4.2
- III. This can make the actual maintenance testing difficult not the impact analysis task.
- IV. If the people involved in impact analysis are new to the project or do not possess domain knowledge it can make the impact analysis difficult. Syllabus 2.4.2

Thus:

- a) Not correct
- b) Not correct
- c) **Correct**
- d) Not correct

Question 10

Which of the following statements (1-4) are related to different test type (A-D)?

1. Test design and execution requires knowledge of changes made to the system.
 2. Test design and execution requires knowledge of how the code is built and how the data is stored.
 3. Test design and execution requires knowledge of the business domain of the system.
 4. Test design and execution requires knowledge of the inherent drawbacks of design or technology.
- A. Functional testing.
 - B. Non-functional testing.
 - C. White-box testing.
 - D. Change-related testing.
- a) 1D, 2B, 3A, 4C
 - b) 1D, 2C, 3A, 4B**
 - c) 1C, 2B, 3A, 4D
 - d) 1D, 2A, 3B, 4C

FL-2.3.1 (K2) Compare functional, non-functional, and white-box testing

Justification

1. For **change related testing** thorough knowledge of changes made to the system is required. (Syllabus 2.3.4) **(1D)**
2. For **White-box test design and execution** knowledge of the way the code is built and how data is stored is required. (Syllabus 2.3.3) **(2C)**
3. For **Functional test design and execution** knowledge of the business, domain is required. (Syllabus 2.3.1) **(3A)**
4. For **Non-functional test design and execution** knowledge of the inherent drawbacks of design or technology is required. (Syllabus 2.3.2) **(4B)**

Thus:

- a) Not correct
- b) Correct**
- c) Not correct
- d) Not correct

Question 11

Which of the following statements is a characteristic of good testing and applies to any software development life cycle model?

- I. All the test levels are completed for each developed feature.
 - II. Each test level in the development lifecycle has a test objective specific to that level.
 - III. Test analysis and design for a given test level begins after the corresponding development activity is completed.
 - IV. Testers are involved in discussion and review of all the work products as soon as the drafts are available.
- a) I & II
 - b) III & IV
 - c) **II & IV**
 - d) II & III

FL-2.1.1 (K2) Explain the relationship between software development activities and test activities in the software life cycle

Justification

- I. This is false, it is not necessary that all the test levels are required in all situations. For example: when the system under test is not integrated with other systems, system integration testing is not required.
- II. This is true, each test level should have test objectives specific to that level (Syllabus 2.1.1)
- III. This is false, Test analysis and design for any given test level should begin **during** the corresponding development activity.
- IV. This is true, Testers should participate in discussions to define and refine requirements and design, and should be involved in reviewing work products as soon as drafts are available. (Syllabus 2.1.1)

Thus:

- a) Not correct
- b) Not correct
- c) **Correct**
- d) Not correct

Question 12

How can white box tests be applied during component testing?

- a) By verifying that all work process flows are covered.
- b) **By verifying that statement coverage and code decision coverage for all paths in the code is achieved.**
- c) By verifying that all web page navigations are covered.
- d) By verifying that large volumes of data can be successfully transferred between components.

FL-2.3.2 (K1) Recognize that functional, non-functional and white-box tests occur at any test level

Justification

- a) Not correct – Relevant for acceptance testing.
- b) **Correct** –For component testing, tests are designed to achieve complete statement and decision coverage for all components that perform financial calculations. (Syllabus 2.3.5)
- c) Not correct – Relevant for system testing.
- d) Not correct – Relevant for component integration testing.

Question 13

Which of the following work products (1-4) are commonly used as a test basis for the different level of testing (A-D)?

1. Installation procedures
2. Data models
3. Workflows
4. State diagrams

- A. Component testing
- B. Integration testing
- C. System testing
- D. Acceptance testing

- a) 1D, 2A, 3B, 4C.
- b) 1D, 2B, 3A, 4C.
- c) 1C, 2D, 3B, 4A.
- d) 1A, 2C, 3D, 4B.

FL-2.2.1 (K2) Compare the different test levels from the perspective of objectives, test basis, test objects, typical defects and failures, and approaches and responsibilities

Justification

- **Installation procedures** are typical work products used as test basis for **Operational acceptance testing (OAT)**. Syllabus 2.2.4. **(1D)**
- **Data models** are typical work products used as test basis for **component testing**. Syllabus 2.2.1. **(2A)**
- **Workflows** are typical work products used as test basis for **integration testing**. Syllabus 2.2.2. **(3B)**
- **State diagrams** are typical work products used as test basis for **system testing**. Syllabus 2.2.3. **(4C)**

Thus:

- a) **Correct**
- b) Not correct
- c) Not correct
- d) Not correct

Question 14

For the following situation (1-4) below, choose the BEST suited review types (A-D)?

1. Review of the Test plan document.
2. Review of the design document for a safety critical system.
3. Review of peer test cases.
4. Review of Technical work products for a complex system.

- A. Technical Review.
- B. Informal Review.
- C. Inspection.
- D. Walkthrough.

- a) 1D, 2C, 3B, 4A.
- b) 1B, 2C, 3D, 4A.
- c) 1D, 2B, 3C, 4A.
- d) 1B, 2D, 3C, 4A.

FL-3.2.3 (K2) Explain the differences between different review types: informal review, walkthrough, technical review and inspection

Justification

1. **Walkthrough** is most suitable for the review of the Test plan document. (Syllabus 3.2.3) **(1D)**
2. **Inspection** is best suited for review of design documents for a safety critical system (Syllabus 3.2.3) **(2C)**
3. **Informal Review** or pair review is best suited for review of peer test cases (Syllabus 3.2.3) **(3B)**
4. **Technical Review** is best suited for review of a technical document. (Syllabus 3.2.3) **(4A)**

Thus:

- a) **Correct**
- b) Not correct
- c) Not correct
- d) Not correct

Question 15

A mobile company is developing software for its new series of phones. The detailed specifications are captured in the form of use-cases. These use cases have detailed descriptions of how mobile phone users use the mobile phone for the following flow of events:

- Making phone call
- Receiving phone call
- Viewing call history
- Sending SMS
- Maintaining contacts

The team is using perspective-based review techniques to review these use-cases. For the reviewer who has taken the role of “tester”, which of the following will be appropriate to check the effectiveness of the work-product?

- a) Try to find the defects in the work product from a tester’s perspective.
- b) Find the issues in the work product based on the potential defects list.
- c) **Generate a draft of end-to-end test cases using the work product.**
- d) Review the work product sequentially, identifying and documenting the issues as they are encountered.

FL-3.2.4 (K3) Apply a review technique to a work product to find defects

Justification

- a) Not correct – This is role-based review and not a perspective-based review.
- b) Not correct – This is checklist-based review and not a perspective-based review.
- c) **Correct** – In perspective-based review, reviewer attempts to use the work product under review to generate the product they would derive from it. In this situation, the tester will try to create a draft of end-to-end test cases from the use-cases to see if all the necessary information is included. (Syllabus 3.2.4)
- d) Not correct – This is an ad hoc review, not a perspective-based review.

Question 16

Which of the following TWO statements about static testing are true?

- a) **Static testing is an important part of security testing.**
- b) Static testing decreases the development cost but increases the testing cost.
- c) **Static testing increases development productivity.**
- d) Static testing can find run-time issues before dynamic testing.

FL-3.1.2 (K2) Use examples to describe the value of static testing

Justification

- a) **Correct** – This is true. Static analysis is an important part of security testing. (Syllabus 3.1)
- b) Not correct – This is not completely true. Static testing decreases both the development and testing cost. (Syllabus 3.1.2)
- c) **Correct** – This is true. Static testing increases development productivity (e.g., due to improved design, more maintainable code). (Syllabus 3.1.2)
- d) Not correct – This is false. Run time issues can be found only by dynamic testing. (Syllabus 3.1.3).

Question 17

In the formal review, who is primarily responsible for collating and recording potential defects during the review activity?

- a) Facilitator
- b) Reviewers
- c) Author
- d) **Scribe**

FL-3.2.2 (K1) Recognize the different roles and responsibilities in a formal review

Justification

- a) Not correct- It is not the responsibility of Facilitator.
- b) Not correct- It is not the responsibility of Reviewers.
- c) Not correct- It is not the responsibility of the Author.
- d) **Correct** – It is the responsibility of the **Scribe**. (Syllabus 3.2.2)

Question 18

Which of the following activities are related to “Initiate review” activity of the work product review process?

- I. Selecting the participants and allocating roles.
 - II. Distributing the work product and supporting material to the participants.
 - III. Estimate effort and timeframe.
 - IV. Review of the work product.
- a) I and II
 - b) II and III
 - c) II, III and IV
 - d) **Only II**

FL-3.2.1 (K2) Summarize the activities of the work product review process

Justification

- I. Selecting the participants and allocating roles is done during **Planning**. (Syllabus 3.2.1)
- II. Distribution of the work product and supporting material (issue log forms, checklist etc.) to the participants is done during **Initiate review** activity. (Syllabus 3.2.1)
- III. Estimate effort and timeframe activity are done during **Planning**. (Syllabus 3.2.1)
- IV. Review of the work product is done during **Individual review**. (Syllabus 3.2.1)

Thus:

- a) Not correct
- b) Not correct
- c) Not correct
- d) **Correct**

Question 19

A car park system calculates the parking fee based on the total time the car is in the parking area. The total time is calculated as the difference between the entry and exit time.

- If the total time is less than or equal to 8 hours the fee is \$11
- For total time more than 8 hours up to 20 hours the fee is \$28
- After that, for each 2-hour slot, \$5 is added to the fee.

The maximum time for which the passenger can be charged is 24 hours and time is measured to the nearest minute.

Which of the following are a valid set of values (in hrs.mins) identified by two-point boundary value analysis?

- a) 0, 1, 8, 9, 19, 20, 21, 22, 23, 24
- b) 0.0, 0.1, 8.01, 20.00, 20.01, 22.00, 23.59
- c) 7.59, 8.00, 8.01, 19.59, 20.00, 20.01, 22.00, 22.01
- d) **8.00, 8.01, 20.00, 20.01, 22.00, 22.01, 24.00, 24.01**

FL-4.2.2 (K3) Apply boundary value analysis to derive test cases from given requirements

Justification

The following partitions can be identified:

1. – 8 Two-point boundaries 8.00,8.01
2. 8.01 – 20 Two-point boundaries 8.00, 8.01, 20.00,20.01
3. 20.01 – 22 Two-point boundaries 20.00, 20.01, 22.00, 22.01
4. 22.01 – 24 Two-point boundaries 22.00, 22.01, 24.00, 24.01

Thus:

- a) Not correct –These are not valid values, time is measured to the nearest minute.
- b) Not correct – Does not include valid two-point boundary values. There are also values not necessary for two-point boundary value analysis
- c) Not correct – Does not include valid two-point boundary values. There are also values not necessary for two-point boundary value analysis.
- d) **Correct** –Includes all valid two-point boundary values.

Question 20

Consider the following statement:

“The code contains only a single IF statement with an explicit true statement and no explicit false statement.”

Which of the following is correct with respect to the above scenario?

- a) For this scenario, the number of tests required for 100% statement coverage should be equal to the tests required for 100% decision coverage.
- b) For this scenario, the number of tests required for 100% statement coverage should be less than the tests required for 100% decision coverage.**
- c) For this scenario, a number of tests required for 100% statement coverage should be more than the tests required for 100% decision coverage.
- d) For this scenario, additional information is required to get the exact number of tests to achieve 100% statement and 100% decision coverage.

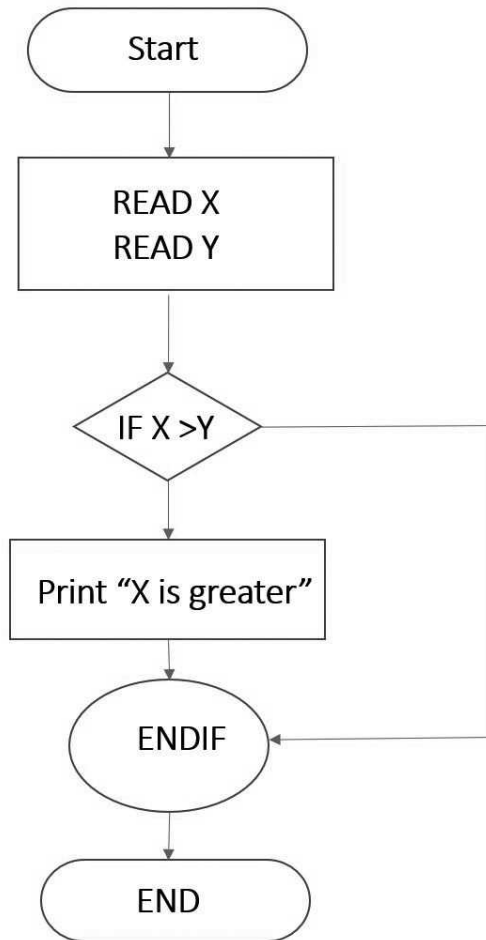
FL-4.3.1 (K2) Explain statement coverage

Justification

- a) Not correct – See explanation below.
- b) Correct** – For this scenario, the tests required for 100% statement coverage are less than the tests required for 100% decision coverage. See explanation below.
- c) Not correct – This is never possible, number of test cases required to achieve 100% statement coverage will be always less or equal to the tests required to achieve 100% decision coverage.
- d) Not correct – There is enough information, see explanation below.

Following code has one decision with an explicit true statement and no explicit false statement:

1. DEC X, Y integer
2. READ X, READ Y
3. IF X> Y THEN
4. “X is greater than Y”
5. END IF

Flowchart for this code:

From the flow chart, it is clear that one test is required to achieve 100% statement coverage and two tests are required to achieve 100% decision coverage.

For the first test, the values are $X=100$, $Y=50$

For the second test, the values are $X=50$, $Y=100$

The first test will give 100% statement coverage as all the executable statements in the code are executed but not all the decision, therefore, another test is required to test the false outcome of the decision.

Thus, the tests required for 100% statement coverage are less than the tests required for 100% decision coverage.

Question 21

Which one of the following is the BEST description of decision coverage?

- a) It is measured as all the true decision outcomes in the source code divided by the total number of decision outcomes in the code expressed as a percentage.
- b) **It is measured as the number of decision outcomes executed by the tests divided by the total number of decision outcomes in the code, expressed as a percentage.**
- c) It is measured as the number of decision outcomes of the source code executed by test cases that are passed, expressed as a percentage.
- d) It is measured as the number of decision outcomes divided by the statement coverage, expressed as a percentage.

FL-4.3.2 (K2) Explain decision coverage**Justification**

- a) Not correct – It is not only the true decision but both true and false decisions which are required to get the coverage.
- b) **Correct** – Decision testing exercises all the decisions in the code. Decision coverage is measured as the number of decision outcomes executed by the tests divided by the total number of decision outcomes in the code. It is normally expressed as a percentage. (Syllabus 4.3.2)
- c) Not correct – Decision coverage is not related to test execution results.
- d) Not correct – Statement coverage has no impact on decision coverage.

Question 22

Which of the following statements describing statement coverage and decision coverage are true?

- I. 100% statement coverage does not ensure that all decisions are tested whereas 100% decision coverage ensures all statements are tested.
- II. Statement testing may provide less coverage than decision testing.
- III. Statement coverage finds defects in code that was not exercised by other tests and decision coverage finds defects in the code where other tests have not taken into account the false outcomes.
- IV. Statement testing and decision testing will provide the same coverage when testing a code having an IF statement without an ELSE statement.

- a) **I and II.**
- b) I, II and III.
- c) I, II and IV.
- d) I, II, III and IV.

FL-4.3.3 (K2) Explain the value of statement and decision coverage

Justification

- I. This is true. Achieving 100% decision coverage guarantees 100% statement coverage (but not vice versa). (Syllabus 4.3.3)
- II. This is true. Statement testing provides less coverage than decision testing. (Syllabus 4.3.3)
- III. This is false. The first part is correct but the second part is not correct. Decision coverage finds defects in the code where other tests have not taken into account both **true** and false outcomes. (Syllabus 4.3.3)
- IV. This is false. Even if there is not ELSE statement the false outcome still needs to be verified, decision testing will ensure that it is tested. (Syllabus 4.3.3)

Thus:

- a) **Correct**
- b) Not correct
- c) Not correct
- d) Not correct

Question 23

Which of the following BEST defines the coverage?

- a) The degree to which a component or system can be used by specified users.
- b) **The degree to which specified coverage items have been exercised by a test suite expressed as a percentage.**
- c) The degree of effectiveness and efficiency with which tests can be designed and executed for a system.
- d) The degree to which a relationship can be established between two or more work products.

FL-4.x (K1) Keywords**Justification**

- a) Not correct – This is the definition of usability from the Glossary.
- b) **Correct** – This is the definition of **coverage** from the Glossary.
- c) Not correct – This is the definition of testability from the Glossary.
- d) Not correct – This is the definition of traceability from the Glossary.

Question 24

A new online system is developed to generate a weekly train pass for different categories of commuters. To buy a weekly pass, the user must select only one of the option from the following category:

- Office staff
- Students
- Foreign tourists

The following decision table has been designed to test the logic and decide whether the user qualifies for the discount.

	R1	R2	R3	R4	R5	R6	R7	R8
Conditions								
Office staff	Y	N	Y	N	Y	N	Y	N
Students	N	N	Y	Y	N	N	Y	Y
Foreign tourists	N	N	N	N	Y	Y	Y	Y
Actions								
Discount	N	N	N	Y	N	Y	N	N

Which of the test cases wouldn't occur in a real situation and can be eliminated from the decision table?

- a) R1, R2, R3, R5, R7, and R8.
- b) R2.
- c) **R2, R3, R5, R7, and R8.**
- d) R3, R5, R7 and R8.

Select one option.

FL-4.2.3 (K3) Apply decision table testing to derive test cases from given requirements

Justification

As mentioned earlier, the user can select only one category so all the test cases where the user has selected multiple categories (R3, R5, R7, R8) are eliminated. Also, the category selection is a must so R2 is not possible. So R2, R3, R5, R7, and R8 are logically impossible.

Hence

- a) Not correct.
- b) Not correct.
- c) **Correct.**
- d) Not correct.

Question 25

Which of the following can be used for creating test conditions, test cases, and test data for white-box testing techniques?

- I. Business Processes
- II. Distribution of likely defects
- III. Detailed design of the system
- IV. Code of the test object

- a) I & II.
- b) III & IV.**
- c) I & III.
- d) II & IV.

FL-4.1.1 (K2) Explain the characteristics, commonalities, and differences between black-box test techniques, white-box test techniques and experience-based test techniques

Justification

- I. Business Processes are used for creating test conditions, test cases, and test data for **black-box testing techniques**. (Syllabus 4.1.2)
- II. Distribution of likely defects is used for creating test conditions, test cases, and test data for **experience-based testing techniques**. (Syllabus 4.1.2)
- III. Detailed design of the system is used for creating test conditions, test cases, and test data for **white-box testing techniques**. (Syllabus 4.1.2)
- IV. Code of the test object is used for creating test conditions, test cases, and test data **for white-box testing techniques**. (Syllabus 4.1.2)

Thus:

- a) Not correct
- b) Correct**
- c) Not correct
- d) Not correct

Question 26

An online train reservation system calculates ticket concession based on the passenger's age. The age can only be entered as a positive number, without any decimal or fractional parts.

- For passenger aged, 5 or less the concession is 100%.
- For passenger aged over 5 and up to 18, the concession is 60%.
- For passenger aged over 18 and up to 60, there is no concession.
- For a passenger aged more than 60, the concession is 50%.

Which of the following is the most likely set of values (in years) identified by two-point boundary value analysis?

- a) 4, 5, 6, 17, 18, 19, 59, 60, 61
- b) 5, 18, 60
- c) -1, 0, 5, 6, 18, 19, 60
- d) **5, 6, 18, 19, 60, 61**

FL-4.2.2 (K3) Apply boundary value analysis to derive test cases from given requirements

Justification

The following partitions can be identified:

1. – 5 Two-point boundaries 5, 6
2. 6 – 18 Two-point boundaries 5, 6, 18, 19
3. 19 – 60 Two-point boundaries 18, 19, 60, 61
4. 61 – Two-point boundaries 60, 61

Thus:

- a) Not correct – These are values for three-point boundary value analysis, not two-point analysis.
- b) Not correct – Does not include all two-point boundary values.
- c) Not correct – Does not include all two-point boundary values. There are also values not necessary for two-point boundary value analysis.
- d) **Correct** – Includes all two-point boundary values.

Question 27

Which of the following TWO statements clearly describe the difference between exploratory testing and error guessing?

- a) Exploratory testing is applicable only for system level, whereas error guessing is applicable for all test levels.
- b) **In Exploratory testing, tests are not pre-defined, whereas in error guessing the tests are pre-defined.**
- c) **In Exploratory testing test results are used to create additional tests for areas that need more testing, whereas in error guessing previous defect list can be used to create additional tests for areas that need more testing.**
- d) Exploratory testing requires testers to be experienced in similar applications, whereas for error guessing technique testers experience is not important.

FL-4.4.2 (K2) Explain exploratory testing**Justification**

- a) Not correct – Both can be applied to all levels of testing.
- b) **Correct** – In exploratory testing, tests are not pre-defined but designed, executed, logged, and evaluated dynamically during test execution. In error guessing tests are pre-defined based on the list of possible mistakes, defects, and failures. (Syllabus 4.4.2/4.4.1)
- c) **Correct** -In exploratory testing, the test results are used to learn more about the component or system and to create tests for areas that may need more testing. In error guessing, possible mistake, defect or failure lists can be used to create tests for areas that may need more testing. (Syllabus 4.4.2/4.4.1)
- d) Not correct – Both are part of experience-based test techniques and require experienced testers.

Question 28

The system for income tax department calculates the tax based on the individual income earned during the financial year. Income up to \$20,000 is exempted from tax, thereafter tax is calculated based on the following rules:

- For income up to \$45,000 the tax is \$5672 plus 4% of the amount above \$20,000
- For income more than \$45,000 and up to \$75,000 the tax is \$6670 plus 6% of the amount above \$45,000
- For income more than \$75,000 and up to \$120,000 the tax is \$9628 plus 8% of the amount above \$75,000
- For income more than \$120,000 the tax is \$19,672 plus 10% of the amount above \$120,000

Which of these groups of numbers which cover all the valid equivalence classes?

- a) (\$0), (\$20,000), (\$45,000), (\$75,000), (\$120,000)
- b) (\$20,000), (\$45,000), (\$75,000), (\$120,000), (\$150,000)**
- c) (\$20,000), (\$45,000), (\$75,000), (\$120,000)
- d) (\$45,000), (\$75,000), (\$120,000), (\$150,000)

FL-4.2.1 (K3) Apply equivalence partitioning to derive test cases from given requirements

Justification

Valid equivalence partitions are below:

1. $0 < \text{income} \leq 20,000$
2. $20,000 < \text{income} \leq 45,000$
3. $45,000 < \text{income} \leq 75,000$
4. $75,000 < \text{income} \leq 120,000$
5. $120,000 < \text{income}$.

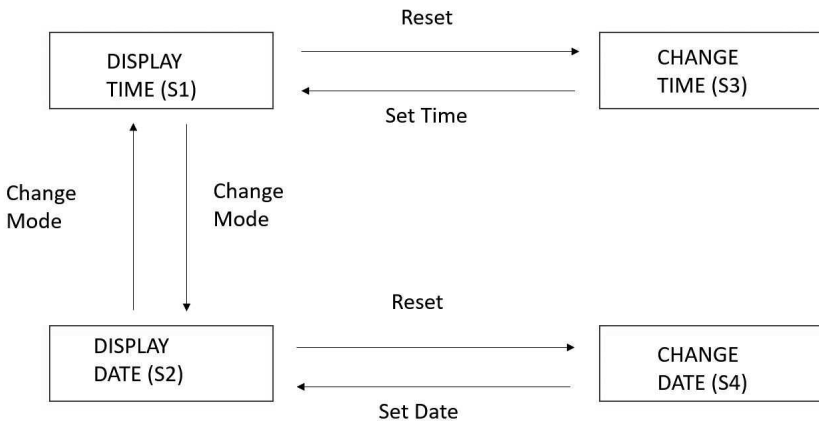
(\$20,000), (\$45,000), (\$75,000), (\$120,000), (\$150,000) covers all the five partitions.

- a) Not correct – see explanation above
- b) Correct** – see explanation above.
- c) Not correct – see explanation above.
- d) Not correct – see explanation above.

Question 29

Consider a system for an electronic device which can display either time or the date based on the user selection.

Following state transition diagram shows the different states and input requests for this device.



How many tests are required to test all the invalid transactions for the above state transition diagram?

- a) 16 tests.
- b) 6 tests.
- c) 9 tests.
- d) 10 tests.**

FL-4.2.4 (K3) Apply state transition testing to derive test cases from given requirements

Justification

A state table can be created for the above state transition diagram. All the states are on the left-hand side of the table, and all the triggers(events) are across the top.

Now to derive the value of each cell of the table input the trigger at the top of the current state.

Example: When the current state is **S1** and '**Change Mode**' trigger is applied the resulting state (end state) is **S2**. This is a valid transaction

because when the trigger is applied the end state is different from the current state.

When the current state is **S1** and ‘**Set Time**’ trigger is applied nothing happens. Therefore, this an invalid transaction or “null” transaction (denoted by ‘-’).

	Change Mode	Reset	Set Time	Set Date
S1	S2	S3	-	-
S2	S1	S4	-	-
S3	-	-	S1	-
S4	-	-	-	S2

From the table, there are 6 valid transactions and 10 invalid or “null” transactions (denoted by ‘-’).

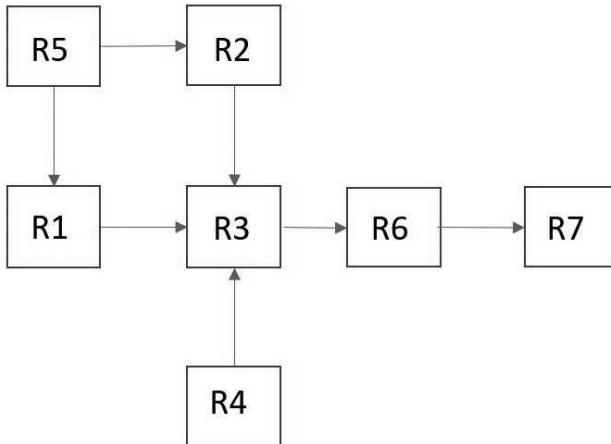
Therefore, test cases for valid transactions =6, Test cases for invalid transactions = 10.

- Not correct – see the explanation above.
- Not correct – see the explanation above.
- Not correct – see the explanation above.
- Correct** – see the explanation above.

Question 30

The following diagram shows the logical dependencies between a set of seven requirements, where a dependency is shown by an arrow.

R2 and R3 are having the highest priority for execution followed by R6. The remaining requirements are having low priority.



If requirement with the highest priority is executed first, which one of the following options structures the test execution schedule?

- a) R2 → R3 → R6 → R5 → R1 → R4 → R7.
- b) R2 → R3 → R5 → R1 → R4 → R3 → R6 → R7.
- c) **R5 → R2 → R1 → R4 → R3 → R6 → R7.**
- d) R5 → R1 → R2 → R4 → R3 → R6 → R7.

FL-5.2.4 (K3) Apply knowledge of prioritization, technical and logical dependencies to schedule test execution for a given set of test cases

Justification

- a) Not correct – everything is dependent on R5, so any test flow that does not start with R5 is FALSE.
- b) Not correct – everything is dependent on R5, so any test flow that does not start with R5 is FALSE.
- c) **Correct** – the tests are specified in a sequence that takes the dependencies into account.

Syllabus 5.2.4: Test cases are run based on their priority levels, usually by executing the test cases with the highest priority first.

However, if the highest priority test cases have dependencies then the dependencies should be considered before priority.

- d) Not correct – R2 is having more priority than R1 so it should be executed before R1.

Question 31

You are testing the software of an electronic watch. This watch can display the current time of different countries based on the selection. During testing, it was observed that for few countries the displayed current time is not accurate.

You have written the following defect report with the following information.

Title: Displayed current time not accurate

Summary: For the country codes (XYZ,...,..) the current time is not displayed correctly in the watch. For the rest of the country codes, the current time is correct.

Conclusion: This may be due to the daylight-saving logic not working correctly.

Which of the following statement about the testers “conclusion” on this defect is correct?

- a) This information is not important as it may confuse the developer who is fixing this problem.
- b) This information is not required as it is the developer’s job to find the root cause of defects.
- c) **This information is important as it may provide additional information required for fixing this problem.**
- d) This information is not part of the standard defect report so does not provide any value in the defect report.

FL-5.6.1 (K3) Write a defect report, covering defects found during testing.

Justification

- a) Not correct – see answer c).
- b) Not correct – see answer c).
- c) **Correct** – Providing conclusions and recommendations by the tester based on their knowledge of domain and application can help the developer to fix the defects. (Syllabus: 5.6)
- d) Not correct – Providing conclusions and recommendations are part of the defect report. (Syllabus: 5.6)

Question 32

Which of the following are potential drawbacks of independent testing?

1. Independent testers will increase the overall project cost.
 2. Independent testers can be biased towards the developers.
 3. Independent testers may be seen as a bottleneck in the project release.
 4. Independent testers may not have all the information needed to perform testing.
-
- a) 1 and 2.
 - b) 1 and 3.
 - c) **3 and 4.**
 - d) 2 and 4.

FL-5.1.1 (K2) Explain the benefits and drawbacks of independent testing

Justification

1. This is not correct. There is more chance that independent testers will find more defects and decrease the project cost.
2. This is not correct. Independent testers will be less biased towards developers.
3. This is a potential drawback that the independent testers may be seen as a bottleneck in the project release. (Syllabus 5.1.1)
4. This is a potential drawback that the independent testers may not have all the information about the test object. (Syllabus 5.1.1)

Thus:

- a) Not correct
- b) Not correct
- c) **Correct**
- d) Not correct

Question 33

Which of the following TWO are included as a part of the test summary report?

- a) Defining the scope and objectives of testing.
- b) Deviations from the test schedule.**
- c) Budgeting of the test activities.
- d) Reusable test work products produced.**

FL-5.3.2 (K2) Summarize the purposes, content, and audiences for test reports

Justification

- a) Not correct – This information has been defined earlier in the test project during planning.
- b) Correct** – This information is included in the test summary report (Syllabus 5.3.2)
- c) Not correct – This information has been defined earlier in the test project during planning.
- d) Correct** – This information is included in the test summary report (Syllabus 5.3.2)

Question 34

Which of the following is the **LEAST** likely one to influence the test effort estimation for a particular project?

- a) The suspension and resumption criteria for testing.**
- b) The complexity of the product domain.
- c) The tools used for testing.
- d) The type and quality of test basis.

FL-5.2.5 (K1) Identify factors that influence the effort related to testing

Justification

- a) Correct** – Suspension and resumption criteria do not affect the test effort estimation.
- b) Not correct – The complexity of the product domain can significantly influence the test effort. (Syllabus 5.2.5)
- c) Not correct – The tools used for testing can significantly influence the test effort. (Syllabus 5.2.5)
- d) Not correct – The type and quality of test basis can significantly influence the test effort. (Syllabus 5.2.5)

Question 35

Which of the following is **NOT** the objective of product risk analysis?

- a) Determine additional activities other than testing to reduce risk.
- b) Determine the test techniques to be used.
- c) Determine the different levels of testing required.
- d) Determine the extent of independent testing required.**

FL-5.5.3 (K2) Describe, by using examples, how product risk analysis may influence the thoroughness and scope of testing

Justification

- a) Not correct – This is one of the objectives of product risk analysis. (Syllabus 5.5.3)
- b) Not correct – This is one of the objectives of product risk analysis. (Syllabus 5.5.3)
- c) Not correct – This is one of the objectives of product risk analysis. (Syllabus 5.5.3)
- d) Correct-** Finding the need and extent of independent testing is NOT the objective of product risk analysis. (Syllabus 5.5.3)

Question 36

Which of the following is the key task for the test manager?

- a) Document deviations from expected results.
- b) Check the status of exit criteria.**
- c) Use appropriate tools to facilitate test process.
- d) Coordinate with network and other teams for environment setup.

FL-5.1.2 (K1) Identify the tasks of a test manager and tester

Justification

- a) Not correct – This is the task of the tester. (Syllabus 5.1.2)
- b) Correct** – This is the task of the test manager. (Syllabus 5.1.2)
- c) Not correct – This is the task of the tester. (Syllabus 5.1.2)
- d) Not correct – This is the task of the tester. (Syllabus 5.1.2)

Question 37

Which of the following factors are MOST likely to be considered while tailoring the test strategy for a project?

- I. Complexity of the project
- II. Product risk analysis
- III. Test estimation technique used
- IV. Goals of the project

- a) I, II, III
- b) I, III, IV
- c) II, III, IV
- d) **I, II, IV**

FL-5.2.2 (K2) Differentiate between various test strategies**Justification**

- I. This is an important factor to be considered while tailoring the test strategy. (Syllabus 5.2.2)
- II. This is an important factor to be considered while tailoring the test strategy. (Syllabus 5.2.2)
- III. This is LEAST likely to be considered while tailoring the test strategy.
- IV. This is an important factor to be considered while tailoring the test strategy. (Syllabus 5.2.2)

Thus:

- a) Not correct
- b) Not correct
- c) Not correct
- d) **Correct**

Question 38

Which of the following TWO can be considered as the objectives of a good configuration management system to support testing activities?

- a) **To ensure a relationship is established between each test items.**
- b) To ensure proper utilization of tools for all test activities.
- c) **To ensure software items are referenced unambiguously in test documentation.**
- d) To ensure all the testing activities follow the test process.

FL-5.4.1 (K2) Summarize how configuration management supports testing

Justification

- a) **Correct** – This is one of the objectives of a good configuration management system. (Syllabus 5.4)
- b) Not correct – Tool utilization is not the purpose of a configuration management system.
- c) **Correct** – This is one of the objectives of a good configuration management system. (Syllabus 5.4)
- d) Not correct – Ensuring adherence to the test process is not the objective of the configuration management system.

Question 39

Which of the following is MOST likely to be considered when selecting a new tool for an organization?

- a) Release cycle of the organization.
- b) The current quality of the test object.
- c) **Training required for the users of new tool.**
- d) The defect density of the projects where the tool can be used.

FL-6.2.1 (K1) Identify the main principles for selecting a tool**Justification**

- a) Not correct – Release cycle of the organization will not affect the new tool selection.
- b) Not correct – Current quality of the test object is not considered for a new tool selection.
- c) **Correct** – Evaluation of the training required is done while looking at the current testing skills of those who will be using the tool. (Syllabus 6.2.1)
- d) Not correct – Defect density of the projects where the tool is used will not affect the tool selection.

Question 40

Which of the following statements related to the purpose of testing tools is True?

- I. Testing tools increase the reliability of testing.
- II. Testing tools can eliminate the need for having some test levels.
- III. Testing tools have a higher level of defect reproducibility.
- IV. Testing tools can automate activities that cannot be done manually.

- a) I, II and III
- b) I, III and IV**
- c) I and IV
- d) I, II, III and IV

FL-6.1.1 (K2) Classify test tools according to their purpose and the test activities they support

Justification

- I. This is true, testing tool can increase the reliability of testing e.g. using tool for large data comparisons. Syllabus 6.1.1
- II. This is false. Testing tools can't eliminate the test levels, they can improve the efficiency of the activities in the test level.
- III. This is true, the testing tools have greater defect reproducibility than manual tests. Syllabus 6.1.1
- IV. This is also true, some of the testing tasks can't be done manually e.g. large-scale performance testing. Syllabus 6.1.1

Thus:

- a) Not correct
- b) Correct**
- c) Not correct
- d) Not correct